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From Guidelines to Questions: Knowledge Graph Construction and Question Generation from Clinical Practice Guidelines

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Zusammenfassung

Klinische Leitlinien geben Empfehlungen für die Diagnose, Behandlung und Überwachung von Patientinnen und Patienten, doch verschiedene Organisationen stellen ähnliche Informationen häufig auf unterschiedliche Weise dar. Sie verwenden unter Umständen abweichende Terminologien, Dokumentstrukturen, Empfehlungsformate und Maßeinheiten. In der Diabetesversorgung kann dies es erschweren, Informationen aus mehreren Leitlinien zusammenzuführen und sie einheitlich für Aufgaben wie die Suche nach klinischen Informationen oder die Generierung von Lernfragen zu nutzen.

Diese Arbeit untersucht, wie Diabetesleitlinien verschiedener Organisationen in einem gemeinsamen Wissensgraphen abgebildet werden können. Ziel ist es, klinische Entitäten, Empfehlungen, numerische Zielwerte und deren Beziehungen in eine strukturierte Form zu überführen, die leitlinienübergreifend abgefragt werden kann. Darüber hinaus wird untersucht, ob der resultierende Graph als kontrollierte Grundlage für die Generierung von Lernfragen dienen kann, deren antwortrelevante Inhalte in den extrahierten Informationen verankert sind.

Das vorgeschlagene System besteht aus vier Stufen: Dokumentenaufnahme, Aufbau des Wissensgraphen, Normalisierung und graphgestützte Fragengenerierung. Es wurde anhand von acht Leitliniendokumenten von vier Organisationen evaluiert, die Typ-1- und Typ-2-Diabetes abdecken. Der resultierende Graph umfasst 2.503 Knoten und 3.485 Kanten über 18 Relationstypen. Die Normalisierung von Entitäten und Maßeinheiten ergab acht leitlinienübergreifende Äquivalenzrelationen zwischen ausgewählten numerischen Zielwerten. Der Durchlauf zur Fragengenerierung mit GPT-4o-mini erzeugte 1.289 Fragen, von denen 1.280 die implementierten Validierungsprüfungen bestanden. Der akzeptierte Katalog enthält 3.048 gespeicherte Umformulierungen und 294 Multiple-Choice-Fragen. Eine separate Stichprobe von 30 Fragen aus einem früheren Teildurchlauf mit Claude Opus 4.8 wurde von zwei Ärztinnen bzw. Ärzten begutachtet; drei eindeutige Fehler ließen sich auf fehlerhafte Relationen im Graphen zurückführen, während eine weitere Frage eher einen Unterschied in der klinischen Praxis als einen sachlichen Fehler aufzeigte.

Die Ergebnisse zeigen, dass der Wissensgraph Informationen aus unterschiedlich strukturierten Diabetesleitlinien zusammenführen und leitlinienübergreifende Abfragen unterstützen kann. Zudem kann der Graph eine kontrolliertere Grundlage für die Generierung von Lernfragen bieten als die direkte Generierung aus dem Leitlinientext. Die Ergebnisse zeigen jedoch auch, dass die Qualität der generierten Fragen von der Qualität des extrahierten Graphen abhängt, insbesondere bei klinischen Relationen. Das Framework ist daher als Grundlage für Anwendungen wie den leitlinienübergreifenden Vergleich, die strukturierte Suche nach klinischen Informationen und die Generierung von Lernfragen gedacht und nicht als Ersatz für die ursprünglichen klinischen Leitlinien.

Abstract

Clinical practice guidelines provide recommendations for the diagnosis, treatment, and monitoring of patients, but different organisations often present similar information in different ways. They may use different terminology, document structures, recommendation formats, and measurement units. In diabetes care, this can make it difficult to combine information from multiple guidelines and use it consistently for tasks such as clinical information retrieval and educational question generation.

This thesis investigates how diabetes guidelines from different organisations can be represented in a common knowledge graph. The aim is to bring clinical entities, recommendations, numerical targets, and their relationships into a structured form that can be queried across guideline sources. The work also examines whether the resulting graph can be used as a controlled basis for generating study questions whose answer-bearing content is grounded in the extracted information.

The proposed system consists of four stages: document ingestion, knowledge graph construction, normalisation, and graph-anchored question generation. It was evaluated using eight guideline documents from four organisations covering Type 1 and Type 2 diabetes. The resulting graph contains 2,503 nodes and 3,485 edges across 18 relation types. Normalisation of entities and measurement units produced eight cross-guideline equivalence relations between selected numerical targets. The GPT-4o-mini question-generation run produced 1,289 questions, of which 1,280 passed the implemented validation checks. The accepted catalogue contains 3,048 stored reformulations and 294 multiple-choice questions. A separate sample of 30 questions, taken from an earlier partial run with Claude Opus 4.8, was reviewed by two medical doctors; three clear errors were traced to incorrect graph relations, while one additional question raised a difference in clinical practice rather than a factual error.

The results show that the knowledge graph can bring together information from differently structured diabetes guidelines and support queries across guideline sources. The graph can also provide a controlled basis for generating educational questions whose factual content is tied to specific graph relations. However, the results also show that the quality of the generated questions depends on the quality of the extracted graph, particularly for clinical relations. The framework is therefore intended as a foundation for applications such as cross-guideline comparison, structured clinical information retrieval, and educational question generation, rather than as a replacement for the original clinical guidelines.

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List of Acronyms

ADA	American Diabetes Association
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CPG	Clinical Practice Guideline
FPG	Fasting Plasma Glucose
GPT	Generative Pre-trained Transformer
GRADE	Grading of Recommendations Assessment, Development and Evaluation
IDF	International Diabetes Federation
IE	Information Extraction
IFCC	International Federation of Clinical Chemistry
JSON	JavaScript Object Notation
KG	Knowledge Graph
LLM	Large Language Model
LPG	Labelled Property Graph
MCQ	Multiple Choice Question
MSTTR	Mean Segmental Type-Token Ratio
NER	Named Entity Recognition
NICE	National Institute for Health and Care Excellence
NLP	Natural Language Processing
OGTT	Oral Glucose Tolerance Test
PDF	Portable Document Format
QA	Question Answering
QG	Question Generation
SGLT	Sodium-Glucose Cotransporter

1. Introduction

1.1. Motivation

Clinical practice guidelines are documents that provide evidence-based recommendations for healthcare professionals. They may describe how to diagnose a condition, which treatments to use, what targets to aim for, and how patients should be monitored. These guidelines are published by professional and national organisations , which often use different document structures, terminology, and measurement units.

This variation makes it difficult to compare guidelines automatically. For example, one organisation may report a fasting plasma glucose threshold in mg/dL, while another reports the same type of measurement in mmol/L. A simple text search can find the numbers, but it cannot determine whether they describe the same clinical concept, convert the units, or preserve which organisation reported each value.

Building a system that compares guidelines automatically is harder than it might first appear, because existing tools for extracting structured information from documents are usually built around one source at a time. A parser or extraction rule written for one organisation's documents often does not work on another organisation's documents, because the two use different layouts, different wording, and different ways of presenting the same kind of information. A system that is only ever tested on documents from a single publisher may work well on those documents while failing on anything else.

This thesis addresses the problem by representing guideline information in a knowledge graph. A knowledge graph is a structured network in which important items, such as organisations, treatments, conditions, and numerical targets, are represented as nodes. Relationships between these items are represented as edges. This structure allows information from different guidelines to be compared and queried systematically.

As a concrete example, consider a clinician or medical educator who wants to know whether the American Diabetes Association (ADA)[1], the National Institute for Health and Care Excellence (NICE)[2], the International Diabetes Federation (IDF)[3], and Diabetes Canada[4] use the same fasting plasma glucose threshold to diagnose diabetes. Working from the source documents, this means opening four guidelines, finding the relevant table or paragraph in each, and converting between mg/dL and mmol/L by hand. In the knowledge graph built in this thesis, the thresholds are already converted to the same unit and linked when they represent the same normalised target, so the question can be

[1] https://www.diabetes.org/
[2] https://www.nice.org.uk/
[3] https://www.idf.org/
[4] https://www.diabetes.ca/

answered with a single query, while each value still points back to the organisation and page that stated it.

The proposed system has four stages. First, it converts PDF guidelines into machine-readable records. Second, it extracts relevant information from those records, using a combination of pattern-based rules and a language model, depending on how predictable the information is. Third, it normalises terms and measurement units so that equivalent information can be compared across organisations. Finally, it uses the resulting graph to generate and validate questions whose factual content is tied to specific, traceable statements from the source guidelines.

The instantiation evaluated in this thesis uses eight documents from four organisations covering Type 1 and Type 2 diabetes. The four-stage architecture is not tied to diabetes, but the specific rules used to read the documents, identify entities, and compare values were built for this corpus; applying the architecture to another domain would require adapting these rules.

1.2. Objectives and Contributions

The main contribution of this thesis is an end-to-end pipeline that converts diabetes clinical guidelines with different formats and terminology into a normalised knowledge graph that can be searched and compared. The graph is also used as a controlled factual source for generating questions. The pipeline converts documents into structured records, extracts relevant information, standardises entities and measurement units, compares information across guidelines, and generates and validates questions from the resulting graph. Its key outcome is that relations between guidelines, such as two organisations using the same diagnostic threshold once units are normalised, become explicit and queryable in the graph, while each statement keeps its source organisation and page. Its key outcome is that relationships between information stated in different guidelines, such as two organisations using the same diagnostic threshold once units are normalised, become explicit and queryable in the graph, while each statement keeps its source organisation and page.

A four-stage pipeline for multi-guideline knowledge graph construction. The pipeline consists of an ingestion stage that converts heterogeneous PDF documents into typed element records, an extraction stage with rule-based and contextual components, a normalisation stage that deduplicates entities and converts units, and a question-generation stage that produces a validated question catalogue from the graph. The stages have defined inputs and outputs, allowing the processing steps to be applied and evaluated separately.

A dual-layer extraction architecture. The extraction stage separates entity types between a rule-based layer and a contextual extraction layer. Rule-based extraction handles types with predictable lexical or structural patterns, including numerical targets and explicitly delimited recommendations, while contextual extraction is intended for drugs, clinical conditions, populations, and interventions that require more contextual interpretation. Each extracted record retains its extraction method for later analysis. The final corpus uses the implemented extraction paths described in Chapter 4. The contextual

extraction path was executed for the four entity categories of drugs, clinical conditions, populations, and interventions.

Cross-guideline equivalence edges created by normalisation. After unit conversion, two numerical targets (for example, two blood glucose thresholds) from different organisations that state the same measurement, comparison, and value are linked by a directed "equivalent" connection. These edges are not directly stated in any single source document. They are created only after independently stated targets have been normalised into a common representation and compared across organisations. In the diabetes corpus, the final graph contains eight such connections, linking diagnostic thresholds across ADA, IDF, and Diabetes Canada.

A graph-anchored question generation and validation pipeline. The question generation stage selects graph edges as anchors and provides the selected graph information as factual context to the language model. The validator then checks the generated question against the graph anchor, including answer consistency, numerical values, and guideline attribution. This provides an automated check that the question is consistent with its graph anchor, although it does not establish independent clinical correctness. Questions derived from cross-organisation equivalence connections draw on information from two guidelines simultaneously, which no single-guideline source in this corpus can support on its own Guo et al. (2024); Bi et al. (2020).

1.3. Thesis Structure

Chapter 2 reviews the background concepts on which this thesis builds: knowledge graphs and graph data models, information extraction including named entity recognition and relation extraction, language model grounding and hallucination, question generation, and clinical practice guidelines.

Chapter 3 defines the four-stage pipeline formally and describes the design decisions behind the schema, the dual-layer extraction architecture, the normalisation functions, and the question generation and validation approach.

Chapter 4 instantiates the pipeline on the diabetes guideline corpus and reports the results: the ingestion statistics, the final knowledge graph, the equivalence edges found by normalisation, the question catalogue, and the outcome of a human clinical review of 30 questions by two medical doctors.

Chapter 5 concludes with a summary of contributions and directions for future work.

2. Background and Related Work

This chapter reviews the concepts and prior work on which this thesis builds. Section 2.1 introduces knowledge graphs and the graph data models used to represent them, together with their use in the biomedical domain. Section 2.2 covers information extraction, the task of identifying entities and relations in text, and the methods used to perform it. Section 2.3 discusses hallucination in large language models and techniques for grounding their output in an external source of knowledge. Section 2.4 reviews automatic question generation and the educational frameworks used to describe question difficulty. Finally, Section 2.5 describes clinical practice guidelines and efforts to represent them in a computable form, and Section 2.6 discusses the work most closely related to this thesis and sets out how the present work differs from it.

2.1. Knowledge Graphs

A knowledge graph organises information as a graph, with nodes standing for entities and edges standing for the relations between them. The term is used in several ways across the literature, and Hogan et al. (2021) note that definitions differ depending on which aspect of the technology is emphasised. For the purposes of this work, a knowledge graph is taken to be a directed graph whose nodes are real-world entities, such as a drug or a clinical condition, and whose labelled edges record how those entities are related. A central property of this representation is that it does not require a fixed tabular schema, so information can be added incrementally and incomplete facts can be represented by leaving the corresponding edges out Hogan et al. (2021). This tolerance for incompleteness is useful when information is assembled from many separate sources that each cover only part of a domain. For example, Figure 2.1 shows a small knowledge graph in which entities such as a dog, cat, rat, and bone are represented as nodes, while labelled edges represent relations between them, such as *chases*, *likes*, and *is a*. The example is deliberately domain-independent and is intended only to illustrate the basic node–edge representation of a knowledge graph. In this thesis, the same principle is applied to clinical guideline concepts, with entities such as drugs, conditions, guidelines, and clinical targets represented as typed nodes. The complete clinical schema used in this work is defined later in Section 3.4.

2.1.1. Graph Data Models

The internal model chosen for a knowledge graph determines how facts and their attributes are stored and queried. Hogan et al. (2021) describe several models, of which two are

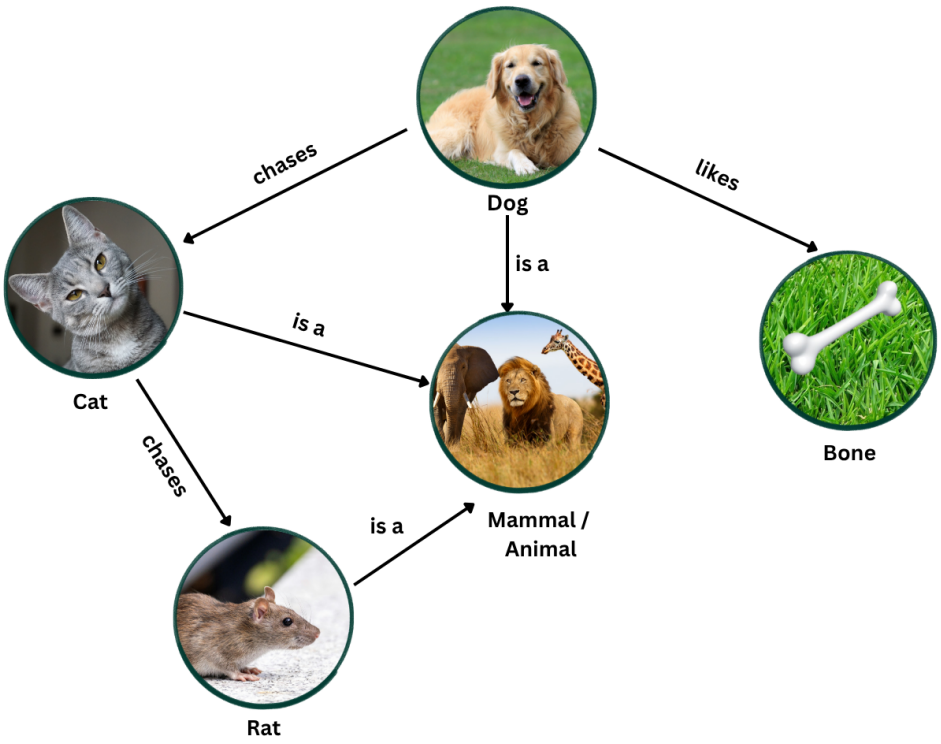


Figure 2.1.: Knowledge Graph Example

relevant to this work: the directed edge-labelled graph, of which the Resource Description Framework (RDF) is the best-known instance, and the property graph.

In a directed edge-labelled graph, every fact is a triple that links a subject node to an object node through a labelled edge, written here as subject–predicate–object. Any additional detail about an entity, such as a numerical value or a source reference, has to be expressed as further triples attached to additional nodes, because an edge in this model carries only its label Hogan et al. (2021). RDF follows this model and adds conventions for naming entities so that graphs published by different parties can be linked together.

A property graph relaxes this restriction. In addition to a type label, each node and each edge may carry an arbitrary set of key-value properties Hogan et al. (2021). A fact such as a recommended dosage or the guideline a statement came from can therefore be stored directly on the node or edge it describes, rather than being broken out into separate triples. Hartig (2014) studies the correspondence between the two models and shows how graphs can be translated between them, which confirms that the choice is one of convenience and expressiveness rather than of what can in principle be represented. The practical consequence is that attributes and edge-level metadata are more compact to express in a property graph, a point returned to when the schema of this work is defined in Chapter 3.

Table 2.1 summarises the principal differences between the two models across the dimensions most relevant to this work.

Feature	Property graph	RDF
Data structure	Typed nodes and edges, each carrying a set of key-value properties	Triples: subject – predicate – object
Edge attributes	Stored directly on the edge as properties	Require intermediate nodes or reification triples
Schema	Optional; the model can be extended at runtime	Ontology-driven; defined using RDFS or OWL
Query language	Cypher (used in Neo4j and Memgraph)	SPARQL
Interoperability	Local identifiers; suited to single-system deployments	Global URIs; supports linked open data across systems
Primary strength	Graph traversal and analytics on highly connected data	Semantic reasoning and cross-system data integration

Table 2.1.: Comparison of property graph and RDF data models.

2.1.2. Knowledge Graphs in the Biomedical Domain

In medicine, knowledge graphs are used to bring together information that is otherwise scattered across terminologies, structured records, and the research literature. A common starting point is a controlled vocabulary that assigns a single identifier to each clinical concept. The Unified Medical Language System integrates many such biomedical vocabularies into one resource, linking multiple names to a single concept and providing tools for extracting concepts from text Bodenreider (2004). Ontologies built on standard coding systems, such as the drug ontology of Hanna et al. (2013) based on RxNorm, serve a similar role by providing stable identifiers that let independently produced data be aligned.

Other work builds biomedical knowledge graphs from clinical text rather than from vocabularies alone. Li et al. (2020) construct a medical knowledge graph from real-world clinical data and apply it to downstream tasks, while Harnoune et al. (2021) use a BERT-based language model to extract clinical entities and relations from biomedical documents for graph construction. These systems generally draw on a single kind of source, most often electronic health records or the biomedical literature. Knowledge graphs built specifically from published clinical guidelines, and in particular from several guidelines issued by different organisations so that their recommendations can be compared, have received far less attention. That setting is the focus of this thesis.

2.2. Information Extraction

Building a knowledge graph from documents requires identifying the entities mentioned in the text and the relations that hold between them. This section describes the task and the main families of methods used to perform it.

2.2.1. Named Entity Recognition and Relation Extraction

The extraction task is usually divided into two parts. Named entity recognition locates mentions of entities in text and classifies them into types, such as a drug or a disease. Relation extraction then determines how those entities are connected, for example that a drug is recommended for a condition Nasar et al. (2021). Together the two tasks turn unstructured text into the typed nodes and edges that a knowledge graph is composed of. Both are long-standing problems in natural language processing, and Nasar et al. (2021) survey the range of methods that have been applied to them.

2.2.2. Rule-Based and Learning-Based Methods

Approaches to information extraction fall broadly into rule-based and learning-based methods. Rule-based extraction relies on hand-written patterns, such as regular expressions or dictionaries of known terms, that match the way information is stated in a particular kind of text.Chiticariu et al. (2013) observe that rule-based information extraction dominates the commercial world even though it is widely regarded as a dead end in academic research, and note that practitioners value rule-based systems for their interpretability. Rules are also a natural fit for text where the relevant facts follow a predictable form, such as a numerical threshold or a graded recommendation, because a small number of patterns can cover most of the cases. Learning-based extraction instead trains a model on annotated examples. Deep learning models, in particular hybrid and joint models for named entity recognition and relation extraction, now represent the state of the art Nasar et al. (2021), and pre-trained language models fine-tuned for relation extraction have outperformed earlier systems on clinical shared tasks Wei et al. (2020). In the clinical domain, Harnoune et al. (2021) use a BERT-based model with a conditional random field layer to extract entities and relations from clinical notes. Shared tasks such as the i2b2 and n2c2 challenges evaluated systems for extracting concepts, assertions, medications, adverse drug events and the relations between them from clinical narratives Uzuner et al. (2011); Henry et al. (2020). Learning-based methods depend on annotated training data, which is scarce for many domains, and understanding the decisions made inside deep architectures remains difficult Nasar et al. (2021).

2.2.3. Hybrid Approaches

Because the two families have complementary strengths, some systems combine them. Chiticariu et al. (2013) argue for rule-based components where precision and control matter, and hybrid designs that pair hand-written rules with machine learning have been used in text-processing tasks to obtain the reliability of rules together with the coverage of learned models Villena et al. (2011). Large language models have more recently been applied to extraction as a further alternative, discussed in the next section. The extraction stage of this work draws on a combination of rule-based and learning-based methods, assigning each to the part of the task for which it is better suited; the details are given in Chapter 3.

2.3. Grounding Language Model Output

Large language models can produce fluent and plausible text, but they are prone to generating statements that are not supported by any real source. This section describes the problem and the techniques used to constrain model output to reliable information.

2.3.1. Hallucination in Language Models

The generation of unsupported or incorrect content by a language model is known as hallucination. Huang et al. (2025) distinguish two forms. Factuality hallucination is when the output contradicts verifiable facts about the world. Faithfulness hallucination is when the output departs from the input or context the model was given, even where it is not wrong in a general sense. They attribute this behaviour in part to the fact that the models are trained to produce coherent and probable text, which does not by itself guarantee that the text is factually correct. In a clinical setting this is a serious concern, because a confidently worded but incorrect statement about a treatment or a threshold could mislead a reader who is not in a position to check it.

2.3.2. Retrieval-Augmented Generation

One established way to reduce hallucination is to ground generation in an external source of knowledge rather than relying on what the model has stored in its parameters. Lewis et al. (2021) introduce retrieval-augmented generation, which combines the parametric memory held in the model's weights with a non-parametric memory in the form of an external document store. At generation time the system retrieves relevant passages and conditions the output on them, so that the generated text is tied to retrieved evidence. The authors report that this grounding improves factual accuracy and provides a way to trace an output back to the source it was drawn from. The same principle underlies a range of later systems that constrain a language model to information supplied at generation time rather than allowing it to draw freely on its parameters.

2.3.3. Schema-Constrained Generation

Providing a model with an explicit, structured input is a strong form of this grounding. Caufield et al. (2024) show that prompting a language model with a defined schema constrains its output to the structure and content requested, which is useful when the goal is to extract or generate information that must conform to a fixed set of types. The question generation stage of this work follows the same idea: rather than asking a language model to produce clinical questions from its own knowledge, it supplies a single fact drawn from the knowledge graph and constrains the model to that fact. Grounding of this kind has been studied mainly for retrieval from text corpora; grounding generation in a structured knowledge graph built from clinical guidelines, so that every generated question is tied to a specific guideline statement, is less commonly addressed. How this is done, and how the output is checked against the supplied fact, is described in Chapter 3.

2.4. Automatic Question Generation

Generating questions automatically from a source of knowledge is an active area of research in natural language processing, motivated by applications in education, information retrieval, and dialogue systems. This section reviews the main approaches and the educational framework used to describe what different kinds of questions demand from a learner.

2.4.1. Approaches to Question Generation

Early question generation systems used rule-based templates that transformed a declarative statement into an interrogative form by pattern substitution. These systems produce grammatical output reliably but are limited to the question types for which templates have been written and lack the flexibility to vary phrasing. Neural approaches have since been developed that learn to generate questions directly from text or from structured knowledge sources. Guo et al. (2024) survey the range of neural methods applied to question generation, covering inputs including free text, knowledge bases, and images. They report that knowledge-base question generation, where the input is a set of structured facts rather than a passage of text, poses particular challenges because the model must convert a formal relation triple into a natural language question that accurately reflects both the entities and the direction of the relationship.

Large language models have also been applied to knowledge-base question generation, for example in few-shot settings where labeled data is scarce. Liang et al. (2023) propose KQG-CoT, which retrieves supportive logical forms from an unlabeled pool and prompts an LLM to build a complex question as a chain of sub-questions. A variant, KQG-CoT+, orders the demonstrations by complexity and improves on the previous few-shot state of the art on PathQuestions. Guo et al. (2024) note in their survey that the central challenge for LLM-based KBQG is designing prompts that steer the model toward the intended question.

Question generation from clinical knowledge has received some specific attention. Nyameino et al. (2020) describe a system that generates questions automatically from a clinical guideline, using the guideline structure to drive the question type and content. The questions produced cover individual recommendations taken from the guideline, rather than comparing recommendations across several guidelines from different organisations.

2.4.2. Multiple-Choice Question Generation

Multiple-choice questions require, in addition to a correct answer, a set of plausible but incorrect options, known as distractors, that a learner who does not know the material might plausibly select. Gao et al. (2019) address distractor generation for reading comprehension questions, using a hierarchical encoder-decoder with attention to generate longer, semantically rich distractors from the article, the question, and the correct option. In a clinical context, plausible distractors are typically other drugs or other thresholds

from the same domain, which makes a knowledge graph a natural resource for distractor selection because it contains the set of all relevant alternatives as named nodes.

2.4.3. Educational Taxonomies and Difficulty

The cognitive demand of a question is typically described using a taxonomy of learning objectives. Anderson and Krathwohl (2001) revised the original taxonomy proposed by Bloom to organise cognitive processes along six levels: remember, understand, apply, analyse, evaluate, and create. The revision shifted the original noun-based category names to verbs in order to emphasise that these are active processes, and it moved the creation category to the highest level to reflect that generating new constructs requires more than evaluation alone. The lower levels, remember and understand, correspond broadly to the ability to recall and interpret factual information, while apply, analyse, and evaluate require the learner to use and reason with that information in increasingly open-ended ways.

In question design, this hierarchy is used to classify the cognitive level a question targets. A question that asks for the value of a specific diagnostic threshold requires recall; a question that presents a clinical scenario and asks which management option is appropriate requires application. Generating questions at a range of levels produces a catalogue that tests different depths of understanding rather than a single level of recall. This work assigns a difficulty level to each generated question based on the relation type that the question was built from, using the Bloom hierarchy as a guide, as described in Chapter 3.

Table 2.2 shows the six cognitive levels of the revised taxonomy together with the kind of question activity associated with each, based on Anderson and Krathwohl (2001).

Process	Associated question activity
Remember	Recall or recognise a specific fact, term, or value
Understand	Explain or restate a concept in one's own words
Apply	Use a known fact or procedure to answer a new question or solve a case
Analyse	Break a scenario into parts and identify how they relate
Evaluate	Judge or justify a clinical decision against given criteria
Create	Design, compose, or propose something novel from existing knowledge

Table 2.2.: Cognitive levels in the revised Bloom taxonomy.

2.5. Clinical Practice Guidelines

Clinical practice guidelines are systematically developed statements that summarise the available clinical evidence on a topic and translate it into recommendations for practice

Nyameino et al. (2020). Their purpose is to improve the quality of care and reduce unjustified variation in practice Peleg (2013). Many guidelines grade the quality of the underlying evidence and the strength of each recommendation, most commonly using the GRADE system Guyatt et al. (2011). In the diabetes domain the organisations whose guidelines are used in this work each produce such statements independently, drawing on their own review processes and addressing their respective healthcare systems and populations.

2.5.1. Evidence Grading and Recommendation Structure

Many guideline producers use a structured grading system to convey the strength of the underlying evidence and the force of each recommendation. One widely used framework is the Grading of Recommendations Assessment, Development and Evaluation system, known as GRADE, which rates the quality of evidence and grades recommendations as strong or weak according to the quality of the evidence and the balance of desirable and undesirable consequences Guyatt et al. (2011). Under GRADE, evidence quality is rated from high to very low depending on study design, consistency of findings, and precision, and recommendations are then graded as strong or conditional based on both the quality of evidence and the balance of benefits against harms. The GRADE system produces recommendations that carry explicit markers of how much confidence a clinician should place in them, which makes the recommendation label a meaningful structured field rather than a narrative qualifier.

In practice, the issuing organisations whose guidelines this work uses do not all follow exactly the same grading system. The ADA uses lettered grades (A through E), NICE uses strength ratings tied to the quality of its evidence review, and Diabetes Canada uses a graded classification from A to D. Regardless of the specific notation, all four organisations distinguish stronger from weaker recommendations in a structured way, and this structure is part of what the ingestion stage of this work captures.

2.5.2. Computer-Interpretable Guidelines

Making guideline content available to computational systems requires moving beyond the PDF and word-processed documents in which guidelines are typically published. Peleg (2013) provides a methodological review of computer-interpretable guidelines, a line of research that formalises clinical guidelines as executable or queryable representations. The review covers modelling languages and acquisition methodologies developed over roughly two decades, and it identifies as a central challenge the gap between the natural language in which clinical guidelines are written and the formal structures that a computer can evaluate. The representations covered by Peleg range from flow-chart-based models to logic-based formalisms, and the work reports that translating a narrative guideline into any of these forms remains time-consuming and requires significant expert input.

Approaches that use artificial intelligence to assist or automate this translation have attracted increasing interest. Large language models can extract structured information from text with little or no task-specific training data Xu et al. (2024); Caufield et al. (2024), which suggests they could also help turn narrative guidelines into structured form.

This thesis takes a related but distinct approach: rather than formalising a guideline into a decision model, it extracts the entities and relations that a guideline states into a property graph, so that the content of several guidelines can be queried and compared in a common structure. The difference is in purpose: computer-interpretable guidelines aim at decision support and care pathway execution, while the knowledge graph constructed in this work aims at information integration and study question generation across multiple guideline sources. Question generation in this setting has so far mainly supported guideline development: Sousa-Pinto et al. (2024) prompted LLMs to propose guideline questions and found that some were not previously raised by the guideline panel.

2.6. Related Work

This section discusses the work most closely related to this thesis and sets out how the present work differs from it. The preceding sections have reviewed individual technical components in isolation; here the focus is on systems that combine several of those components toward a goal comparable to the one pursued here.

Knowledge Graph Construction from Clinical Text Several systems construct biomedical knowledge graphs from clinical text using extraction pipelines that combine named entity recognition with relation extraction. Li et al. (2020) build a medical knowledge graph from real-world clinical data records, and Harnoune et al. (2021) extract clinical knowledge from biomedical documents using a BERT-based model. Both systems work on a single document collection of one type, either electronic health records or biomedical literature, and neither addresses the problem of combining content from multiple separately produced guidelines that each use different structural conventions and different units of measurement. The present work is distinguished by its explicit treatment of structural heterogeneity across sources and its normalisation of measurement units so that numerical thresholds from different guidelines can be directly compared.

Question Generation from Clinical Knowledge Question generation from structured knowledge has developed as a distinct research direction as knowledge graphs have grown richer and more queryable. Bi et al. (2020) generate questions from knowledge bases using explicit type constraints and grammar guidance, showing that the structural properties of a knowledge source can be used to control the factual content and form of generated questions. Guo et al. (2024) survey neural question generation across knowledge bases, text, and images, organize the literature into structured, unstructured, and hybrid NQG, and note that knowledge-base question generation benefits from models that preserve graph structure and mark the roles of heads, relations, and tails rather than treating a subgraph as a flat sequence of triples. Prompting a language model with structured input rather than asking it to generate freely is a common design in knowledge-base question generation Liang et al. (2023), and constraining output to supplied content is also used to reduce unsupported statements Huang et al. (2025).

In the clinical domain specifically, Nyameino et al. (2020) describe a system that generates questions automatically from a single clinical guideline for use in a gamified training

setting. Their work is the closest antecedent to the question generation component of this thesis and establishes that guideline content can be turned into assessment questions automatically.

This thesis extends that line of work in two ways. First, the input is a knowledge graph built from four guidelines rather than a single source document, which makes cross-guideline questions possible. Second, each question is grounded to a specific graph edge selected before the language model is called, and every generated question is validated against that anchor. Questions derived from EQUIVALENT_TO edges, which connect nodes from different organisations that assert the same numerical threshold, require knowledge of how two guidelines compare and are not possible from a single-guideline source.

AI-Assisted Guideline Processing Sousa-Pinto et al. (2024) examine how large language models can support the development of guideline questions, using the ARIA 2024 guidelines as a single case study, and Peleg (2013) reviews the broader effort to produce computer-interpretable representations of individual guidelines for use in decision support. Neither addresses the representation of several guidelines together. The goal of this thesis is different: it is not to execute guidelines as decision logic but to represent the content of several guidelines in a common structure that can answer questions about how those guidelines agree, differ, and complement one another.

3. Knowledge-Graph-based Question Generation Pipeline

This chapter presents a framework for turning a set of clinical guideline documents into a queryable Knowledge Graph, and for using that graph as a controlled source for generating questions. The four-stage architecture is not specific to diabetes, although the schema, extraction rules, canonical entities, and normalisation procedures used in this thesis are configured for the diabetes guideline corpus.

The chapter follows the same order the pipeline runs in. Section 3.2 describes the corpus: which documents are used and why. Section 3.3 presents the document structuring stage, in which each document is parsed into typed element records with page-level provenance. Section 3.4 defines the graph model — the node types, relation types, and formal properties of the graph. Section 3.5 describes the dual-layer approach to identifying entities and relations in the preprocessed records. Section 3.6 explains how extracted records are deduplicated, unit-normalised, and integrated into a single graph. Section 3.7 describes how the finished graph is used as a controlled source for question generation.

3.1. Overview and Objectives

The goal of this work is to build a Knowledge Graph that combines several diabetes guidelines into one structured, queryable representation, and to use that graph to generate questions automatically. Eight documents from four organisations are used, covering diagnosis, screening, glycaemic targets, pharmacological management, and monitoring across both Type 1 and Type 2 diabetes. The pipeline is organised into four stages: ingestion, extraction, normalisation, and question generation. This four-stage architecture is not inherently specific to diabetes, but its implementation in this thesis is configured for the diabetes guideline corpus. This includes publisher-specific document-structuring rules, the Knowledge Graph schema, clinical extraction patterns and canonical entities, unit-normalisation procedures, and the relation-to-question mappings. Applying the same architecture to another clinical domain would therefore require adapting these domain- and corpus-specific components. Diabetes guidelines are used as the test domain mainly because they are structurally messy in different ways depending on the publisher, which meant the pipeline had to be tested against a real spread of formats, not one clean case.

This objective breaks down into three concrete reasons for building the graph in the first place.

Cross-guideline comparison Once each guideline's content is represented as typed nodes and edges, queries can compare positions across organisations, for example by identifying guidelines that specify the same normalised glycaemic target or recommend the same drug as first-line therapy. Numerical targets are kept organisation-specific and are compared only after unit normalisation, as described in Section 3.6.2. The resulting cross-guideline comparisons for the diabetes corpus are reported in Chapter 4.

Structured accessibility of clinical knowledge Guidelines are written for clinicians to read, not for software to parse Peleg (2013). Everything is embedded in prose, numbered lists, and tables, none of which is machine-readable on its own; Section 2.2 discusses this in more depth. Converting that content into a property graph is what makes it possible to query and visualise it, not just search it as text Li et al. (2020); Harnoune et al. (2021).

Question generation for clinical education Once the guideline content sits in a graph, question generation can work directly off graph edges, without needing unstructured passages Guo et al. (2024); Bi et al. (2020). Selected clinical and cross-guideline edges represent statements extracted from or derived from the source guidelines and can be used as anchors for questions at several difficulty levels, which is useful for building learning or assessment material. The question types used here are defined in Chapter 2.

Challenges

The construction of a multi-guideline knowledge graph introduces five challenges that do not arise in a single-source setting.

Challenge 1 is structural heterogeneity. Because the guidelines come from distinct publication workflows, there is no uniform document structure. While some publishers provide clean Unicode text, others introduce extraction artefacts such as missing whitespace. A fixed document-structuring procedure therefore cannot be applied unchanged across the entire corpus. The framework must accommodate publisher-specific structural differences while producing the same typed intermediate representation for downstream processing. The specific repair procedures required for the diabetes corpus are described in Chapter 4.

Challenge 2 is terminological variation. The same clinical idea may be worded differently across organisations. For example "type 2 diabetes" versus "non-insulin-dependent diabetes mellitus" for the same condition. If these forms are kept as separate identities, relations that should meet at one node are split across several nodes. The graph therefore needs a canonical form for such variants before cross-guideline queries are meaningful.

Challenge 3 is unit inconsistency. The guidelines do not consistently use the same measurement units. A numerical comparison is useless if the values are not first put on the same scale. A fasting plasma glucose threshold may be written as 126 mg/dL in one representation and 7.0 mmol/L in another. The numbers look different even though conversion places them at the same threshold. Cross-guideline comparison therefore requires unit normalisation before numerical targets are compared.

Challenge 4 is controlling information the language model introduces beyond what the input actually states. Clinical guideline text is safety-sensitive: an invented drug indication or a fabricated contraindication would misrepresent an actual clinical guideline, which is a more serious failure mode than in general-purpose text generation Huang et al. (2025). The risk is not limited to entity extraction. In question generation, a graph anchor may specify an HbA1c threshold without specifying a patient age. A generated vignette that introduces a numerical age has therefore added information not contained in the anchor. This risk is why the extraction and question-generation stages (Sections 3.5 and 3.7) are built the way they are: the language model is instructed to classify or rephrase information already present in the structured input rather than introduce additional clinical facts, and its output is checked against that input wherever possible.

Challenge 5 is entity scoping. Some concepts should be shared across the whole graph, while others must retain the organisation that stated them. A Drug such as metformin can be represented by one global node even when several guidelines mention it. A numerical target is different. If ADA and IDF both state an FPG threshold of 7.0 mmol/L, the two target nodes remain separate because they represent independent guideline statements. Their agreement is represented by an EQUIVALENT_TO edge rather than by merging the nodes. This distinction is fixed by the schema and carried through deduplication and normalisation.

Contributions

The pipeline consists of four main technical components.

The first is separating entity types into two extraction layers before any extraction runs. This differs from the hybrid approaches of running rules first and letting a language model fill the gaps (or the reverse). Both of those patterns let the two methods work on the same categories, which makes it hard to tell afterwards which method produced which result, or to attribute errors. Here, each entity type is assigned up front to either the rule-based layer, if it has a consistent lexical or typographic pattern, or the language-model layer, if picking it out requires understanding context. Extraction-derived records retain method metadata where applicable, allowing the contribution of each extraction route to be inspected separately.

The second is constraining the language model's extraction output to a fixed schema. The model is given the allowed node and edge types as a prompt-level constraint and is only supposed to classify what is already in the passage, not add to it. This matters more here than in a general information-extraction task, because a model that invents a category or a drug indication produces a graph that looks fine and is wrong in a way a reviewer might not catch.

The third is anchoring every generated question to a specific graph edge before the language model is called. Source relations retain organisation and page provenance where available, while derived relations retain their source context through the graph elements from which they are constructed. When an eligible edge is selected as a question target, the available provenance is retained with the anchor. The model is provided with the

anchor as its factual input and is instructed not to introduce information beyond it. The graph decides what is asked; the model decides how it is phrased.

The fourth is creating cross-guideline equivalence edges after unit normalisation. This works better than trying to extract them from the text directly, since no single guideline document ever states that its own target is equivalent to another organisation's target. For plasma-glucose targets, values expressed in different source units are normalised to mmol/L. Two GlycaemicTarget nodes from different guidelines that share the same metric, operator, and rounded normalised value are then linked with an equivalence edge. This relation only exists because the graph combines multiple guidelines; it could not come out of any single-guideline pipeline.

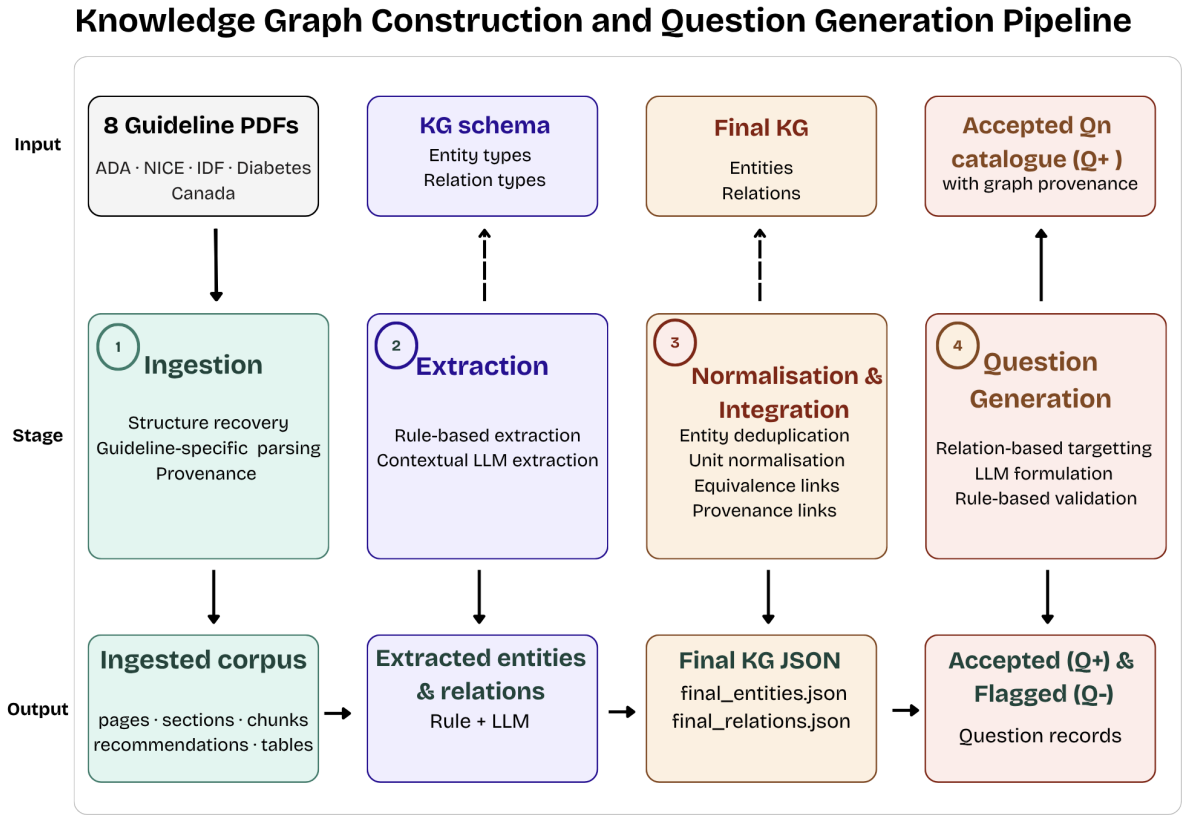


Figure 3.1.: Four-stage knowledge graph construction pipeline.

Figure 3.1 shows the pipeline end to end. Guideline documents go into ingestion, which produces a structured representation of each document's content. Extraction reads that representation and produces two sets of typed entity and relation records, one from the rule-based layer and one from the language-model layer. Normalisation merges those two sets into one graph. Records referring to the same real-world entity are deduplicated into a single node, and numeric thresholds are converted to a common unit so equivalence edges can be detected. Question generation then walks the finished graph, selects eligible edges as anchors, and produces a validated set of clinical questions retaining the provenance available on their source graph relations

3.2. Corpus Design

Building a Knowledge Graph that supports cross-guideline comparison requires a corpus that covers one clinical topic but varies in how that topic is presented. Using documents from a single organisation would produce a clean but narrow graph and defeat the purpose of cross-guideline comparison, so the corpus is selected against four criteria.

Geographic and systemic diversity Guidelines from different healthcare systems reflect different regulatory environments and patient populations. Using organisations from more than one country introduces variation in guideline recommendations and healthcare-system context rather than only the conventions of one national system.

Clinical coverage The selected guideline families need enough overlapping clinical content for cross-guideline comparison to be possible. For this work, the common coverage consists of diagnosis and classification, screening, glycaemic targets, pharmacological management, and monitoring American Diabetes Association Professional Practice Committee for Diabetes (2026); National Institute for Health and Care Excellence (2015b); Federation (2025); Punthakee et al. (2018). A document covering a narrower sub-topic can still be included where it contributes a comparison point on one of these shared areas.

Format diversity Testing the pipeline against documents that all share one layout cannot show whether the pipeline generalises. The corpus requires documents that differ in how recommendations are laid out and in whether they use tables at all, so that ingestion and extraction face varied structure and not one convenient case.

Recommendation specificity Documents need to contain recommendations naming a specific drug, intervention, or measurement threshold, not high-level policy statements with nothing concrete to extract.

The concrete outcome of applying these four criteria, i.e. which documents, how many pages, how many recommendations, is reported in Chapter 4.

3.2.1. Structural Heterogeneity as a Design Challenge

Guideline documents from different organisations are not structurally uniform, and this shapes the design of every stage of the pipeline. Three dimensions of variation are particularly relevant to the pipeline.

Recommendation format

Some guidelines present recommendations as numbered, graded statements that can be located by structural pattern. NICE's Type 2 diabetes guideline uses this format throughout National Institute for Health and Care Excellence (2015b):

> **1.13.1** For adults with type 2 diabetes and no relevant comorbidity, offer modified-release metformin and an SGLT-2 inhibitor. [2026]

The recommendation number, the imperative verb, and the update marker are typographically consistent across the document and can be identified by structural pattern matching. Metformin and SGLT-2 inhibitors are examples of the DRUG entities described in Section 3.4.2.

The ADA Standards of Care also use numbered recommendations, but place them within a multi-column layout alongside commentary and cross-references American Diabetes Association Professional Practice Committee for Diabetes (2026). For example:

> **9.21** In adults with type 2 diabetes without severe hyperglycemia or hyperglycemic crisis, GLP-1-based therapy is preferred to insulin for initial or add-on glucose-lowering therapy. **A**

Although both formats use numbered recommendations, their surrounding document structure differs. The NICE recommendation can be located using its consistent numbering and statement structure, whereas the ADA recommendation must be identified within a three-column layout containing additional commentary and cross-references. This difference affects how the ingestion stage locates and extracts recommendation text.

Tabular content

Some guideline documents present diagnostic criteria in structured tables rather than in numbered statements. This creates a different problem from the numbering-versus-prose distinction above. The issue is not identifying where a recommendation starts and ends, but reconstructing the row-and-column structure before extraction can take place. The IDF and ADA documents present some diagnostic criteria in this form. For example, ADA Table 2.1 includes the following criteria American Diabetes Association Professional Practice Committee for Diabetes (2026):

> **Table 2.1: Criteria for the diagnosis of diabetes in nonpregnant individuals**
>
> FPG (fasting plasma glucose) $\geq$ 126 mg/dL ($\geq$ 7.0 mmol/L). Fasting is defined as no caloric intake for at least 8 h.
>
> *OR*
>
> 2-h PG (plasma glucose measured two hours into an OGTT, an oral glucose tolerance test) $\geq$ 200 mg/dL ($\geq$ 11.1 mmol/L).

When a naive text-stream extractor processes this page without spatial reconstruction, the table structure collapses into a sequence of fragments with no reliable row or cell boundaries. A simplified example of the resulting text stream is shown below.

Listing 3.1: Text-stream output from ADA Table 2.1 before spatial reconstruction.

```
Table 2.1 Criteria for the diagnosis FPG 126 mg/dL 7.0 mmol/L
Fasting defined no caloric intake 8 h 2–h PG 200 mg/dL
11.1 mmol/L OGTT Random plasma glucose 200 mg/dL
```

Pattern matching for an expression such as `FPG ≥ 126 mg/dL` fails on this output because the table structure has collapsed and the comparison operators are not preserved in the extracted text. The association between each metric and its corresponding threshold can therefore no longer be recovered reliably from the text stream alone. Spatial reconstruction of the row-and-column structure is required before rule-based extraction of numerical thresholds from such pages.

The same extraction procedure cannot be assumed to apply to every guideline. NICE NG17, for example, expresses comparable target information in prose rather than in a table. Recommendation 1.6.6 states that adults with Type 1 diabetes should aim for an HbA1c level of 48 mmol/mol (6.5%) or lower (National Institute for Health and Care Excellence, 2015a, recommendation 1.6.6). HbA1c (glycated haemoglobin) is reported here in IFCC units (mmol/mol), which are discussed further below. The ingestion pipeline therefore cannot assume that any single content modality will be present across all guideline families.

Measurement unit conventions

Different organisations report the same class of measurement in different units. Plasma glucose values appear in mg/dL in the ADA document American Diabetes Association Professional Practice Committee for Diabetes (2026) and in mmol/L in the NICE, IDF, and Diabetes Canada documents National Institute for Health and Care Excellence (2015b); Federation (2025); Punthakee et al. (2018). Glycated haemoglobin (HbA1c) appears as a percentage in some documents and in IFCC (International Federation of Clinical Chemistry) units (mmol/mol) in others; the standard method for relating the two scales is given by Hoelzel et al. (2004). Comparing numerical values across guidelines only works after unit normalisation, and only between values of the same kind. Cross-guideline comparison is restricted to targets that share the same metric, operator, and normalised numerical value. The limitations of this criterion are discussed in Chapter 4.

These three dimensions of variation appear across all four guideline families in the corpus, and handling them without excluding a guideline family is a design requirement for the document-structuring and extraction stages described next.

Each of these three problems has been addressed individually in the document-processing literature. PDF text extraction is a recognised difficulty for scientific documents Bast and Korzen (2017); Ramakrishnan et al. (2012), and tools such as Docling combine layout analysis with table structure recognition Auer et al. (2024). Table reconstruction has been studied for scientific articles and document images Zhong et al. (2020); Schreiber et al. (2017). Unit normalisation is needed because clinical measurements are reported on different scales, for example HbA1c in NGSP percent versus IFCC mmol/mol, which are related by a linear regression Hoelzel et al. (2004). Here all three are handled within one pipeline operating across four guideline families at once, without treating any single

family as the default case. The systems surveyed in Section 2.6 address these problems individually, not in combination.

3.3. Structuring Guideline Documents for Extraction

Clinical guideline documents are written for people, not software. Whatever format they arrive in, the actual clinical content is buried in a mix of typographic conventions, layout hierarchy, prose, numbered lists, and embedded tables, and none of that carries explicit markup a program can rely on. This is not really a PDF problem specifically. Human-oriented documents do not necessarily expose the semantic and structural information required by the downstream pipeline in a machine-readable form. Structuring the documents has three requirements. Each piece of content needs a type, its position in the document structure needs to be retained, and anything extracted from it later must remain traceable to the source. The content model below represents these requirements explicitly.

3.3.1. The Document Content Model

Ingestion works against a fixed content model. A guideline document $d \in \mathcal{D}$ is treated as an ordered sequence of content elements $e_1, e_2, \ldots, e_n$, and each element is a triple $e_i = (\tau_i,\ \text{text}_i,\ \pi_i)$, where text_i is the extracted text, π_i is a provenance record (defined below), and τ_i is drawn from a content-type set $\mathcal{T}_C$. $\mathcal{T}_C$ enumerates the structural units a document domain distinguishes, and in general it needs to separate free-form narrative content from any self-contained, explicitly delimited unit the document type uses. The distinction is structural, not semantic. A delimited unit has identifiable boundaries and usually a label the document itself supplies, so it can be located reliably and its identifier kept for provenance. Narrative text has neither. What $\mathcal{T}_C$ actually contains is a domain-specific configuration, not a fixed part of the model. For this corpus,

$$\mathcal{T}_C = \{\textsc{Prose},\ \textsc{Recommendation},\ \textsc{Table},\ \textsc{Section}\}$$

Chapter 4 describes how these four types were derived from the four diabetes guideline documents and gives the extraction rule used for each; a different document domain would likely need a different $\mathcal{T}_C$.

Each element's provenance has two levels. At the element level:

$$\pi_i^{\text{elem}} = (id_i,\ \text{sec_id}_i,\ \text{sec_title}_i,\ p_i^{\text{start}},\ p_i^{\text{end}})$$

id_i is a unique element ID so anything extracted later can be traced back to it; sec_id_i groups elements by section for entity typing; sec_title_i is the actual heading text, which extraction uses to infer relation type; and $p_i^{\text{start}}, p_i^{\text{end}}$ are start and end page numbers, kept because any extracted claim needs to be checkable against the source page it came from. Elements of a self-contained, explicitly delimited type may carry an additional label field where the document itself provides one (e.g. a recommendation number), absent for other types. At the document level, a shared record resolves from id_i:

$$\pi_i^{\text{doc}} = (d_{\text{id}},\ \text{org},\ \text{geo},\ \text{source_file})$$

covering the document ID, issuing organisation, healthcare system, and source filename.

Ingestion itself is a function $\varphi : \mathcal{D} \to 2^{\mathcal{E}}$, and it has to satisfy three things regardless of domain. *Totality*: it produces output for every document, no document is allowed to come out empty just because, say, it lacks one particular content type. *Type coverage*: across the whole corpus, every content type in $\mathcal{T}_C$ actually shows up at least once, so the ingestion logic cannot quietly collapse into handling only one modality. *Provenance completeness*: every element that comes out has a fully populated π_i, so it can be traced back to a source location. Chapter 4 shows a concrete element produced by φ for this corpus and how these three properties were verified against it.

3.3.2. Structural Challenges in Ingestion

PDF extraction is known to introduce artefacts like broken ligatures, lost word boundaries, and collapsed table columns Bast and Korzen (2017); Auer et al. (2024). These are implementation-level engineering problems rather than conceptual ones, but the general requirement they impose on ingestion belongs here: the ingestion stage must detect and repair publisher-specific encoding artefacts before passing text downstream, since a corrupted token boundary breaks any pattern match or rule that assumes intact words. Two such challenges showed up often enough in this corpus to need dedicated handling; Chapter 4 gives the concrete encoding artefact encountered and the publisher-specific repair it required.

The first is extraction fidelity, i.e. documents from different publishers coming out of extraction looking very different internally, in ways that were not spread evenly across the corpus but tied to specific publishers' pipelines. A fix for this kind of artefact has to be written per publisher; one generic normalisation pass does not generalise across document families.

The second challenge is knowing where a recommendation actually ends. Even where the text extracts cleanly, grading scales, evidence summaries, and qualifying clauses sit typographically right next to the recommendation itself, and the boundary between "this is the recommendation" and "this is commentary about it" is not consistent across document families. No single rule covered all four families, so this was configured separately for each.

The complete set of typed, provenance-annotated elements from φ is the only input the extraction stage (Section 3.5) receives — no raw document text passes through beyond this point.

3.4. Knowledge Graph Schema

What the schema allows as a node or an edge determines what the graph can eventually be queried for. This section works through that schema in order: the formal graph model, then the node types, then the relation types, and finally why these particular types were chosen over the alternatives that were considered and dropped.

3.4.1. Knowledge Graph Model

The model used here is the labelled property graph, where nodes and edges are typed and can carry key-value properties Hogan et al. (2021); Hartig (2014). Formally, the graph is

$$G = (V,\ E,\ \mathcal{T}_V,\ \mathcal{T}_E,\ \phi,\ \psi,\ \lambda) \tag{3.1}$$

where $\mathcal{T}_V$ is the node type set (Section 3.4.2) and $\mathcal{T}_E$ is the relation type set (Section 3.4.3), defined before the rest of the tuple since both are used in what follows. V is the set of nodes, and $E \subseteq V \times \mathcal{T}_E \times V$ is the set of directed typed edges, each edge $e = (v_i, r, v_j)$ connecting a source node to a destination node via a relation type $r \in \mathcal{T}_E$. $\phi : V \to \mathcal{T}_V$ assigns every node exactly one type, and $\psi : E \to \mathcal{T}_E$ assigns every edge exactly one relation type. $\lambda : V \cup E \to 2^{\mathcal{P}}$ maps nodes and edges to key-value properties; this is where clinical attributes and provenance metadata are stored.

Two things about the underlying requirements pushed this toward a property graph over RDF. In general, different entity types need different attribute sets: in this corpus, for example, a GlycaemicTarget node needs a metric name, an operator, a value, and a unit, while a Drug node needs none of those. A property graph handles this directly through λ. RDF would need a separate triple per attribute, splitting one clinical fact across several statements Barrasa and Webber (2023). Provenance was the second factor. Clinical relationships need to remain traceable to their source context, so provenance must be representable on edges as well as on nodes. Property graphs support edge properties directly. RDF needs named-graph reification or the singleton property pattern to attach metadata to a statement, which adds structural overhead without a corresponding benefit for this use case.

The corpus $\mathcal{D} = \{d_1, d_2, \ldots, d_n\}$ is the only factual source for G. Building G means applying the extraction function ξ (Section 3.5) to the output of φ, then applying the normalisation function η (Section 3.6) to merge the result into a deduplicated instance of G.

3.4.2. Node Types and Their Roles

The node type set is:

$$\begin{aligned}\mathcal{T}_V = \{&\textsc{Organisation},\ \textsc{Guideline},\ \textsc{Condition},\ \textsc{Drug},\ \textsc{ClinicalCondition},\\ &\textsc{Intervention},\ \textsc{Population},\ \textsc{GlycaemicTarget},\ \textsc{Recommendation},\\ &\textsc{MonitoringProtocol},\ \textsc{Table},\ \textsc{Test},\\ &\textsc{TreatmentRecommendation}\}\end{aligned}$$

These thirteen types came from inspecting the four guideline families in the corpus American Diabetes Association Professional Practice Committee for Diabetes (2026); National Institute for Health and Care Excellence (2015b); Federation (2025); Punthakee et al. (2018), not from an existing medical ontology, and no claim is made that the same thirteen types would work in another clinical domain. An initial list was drawn from the recurring subjects of the recommendation statements, then went through two rounds of

manual review. Types that could not be reliably populated from the corpus were dropped, and types that were quietly conflating two different subjects were split apart. The thirteen types split across three categories: global types, organisation-scoped types, and structural types that sit outside both scoping levels.

Global node types

A drug is the same drug no matter which guideline mentions it, and a clinical condition is the same condition no matter which body defines it. These concepts are represented as global nodes, independent of any one guideline, and that independence is what makes cross-guideline comparison possible in the first place. A single global DRUG node can participate in FIRST_LINE_FOR relations originating from several guidelines, allowing recommendations about the same drug to be compared directly. The global types are DRUG, CONDITION, CLINICALCONDITION, INTERVENTION, and TEST.

Organisation-scoped node types

A glycaemic target value stated in one guideline is not automatically the same statement as a numerically similar value in another. Each is issued by a specific organisation, for a specific patient population, and merging the two on numerical similarity alone would erase that distinction. Organisation-scoped nodes keep each guideline's own position separate by carrying the issuing body as a required property. The organisation-scoped types are GLYCAEMICTARGET, RECOMMENDATION, TREATMENTRECOMMENDATION, MONITORINGPROTOCOL, and TABLE.

Structural node types

Three types are structural, not clinical. ORGANISATION nodes represent the guideline-issuing bodies and sit at the root of the provenance hierarchy. GUIDELINE nodes represent individual published documents and connect to their issuing organisation by a PUBLISHES edge. Where a direct structural relation exists, such as HAS_RECOMMENDATION or CONTAINS_TABLE, the extracted node is linked directly to its source guideline; other organisation-scoped nodes retain guideline-level provenance through organisation and source metadata rather than a direct edge. POPULATION nodes represent patient subgroups, letting the graph distinguish a recommendation for all adults with Type 2 diabetes from one aimed at a more narrowly defined subgroup. The corpus contains many such subgroups stated at different levels of specificity, for example distinctions by age, by pregnancy status, or by a named comorbidity. Population is treated as structural, not global, because these subgroup descriptions are worded differently across organisations and in some cases across recommendations from the same organisation; in the corpus, a subgroup only carries meaning in relation to the specific recommendation and guideline that names it. A small example shows why Population is kept structural. The population descriptions are extracted contextually rather than by fixed pattern. ADA recommendations reference an older-adult subgroup using several different phrasings across different recommendations, including "older adults with type 2 diabetes," "older adults," and "indi-

viduals aged 70 years and older." These are not treated as one canonical Population node the way DRUG mentions of metformin are, because merging them would erase which specific recommendation each phrasing came from, and because the boundary of "older" is not stated consistently even within a single guideline.

3.4.3. Relation Types and Their Semantics

The relation type set is:

$$\mathcal{T}_E = \{\ \text{PUBLISHES},\ \text{HAS__RECOMMENDATION},\ \text{CONTAINS__TABLE},\ \text{FIRST__LINE__FOR},\ \text{SECOND__LINE__FOR},\ \text{RECOMMENDED__FOR},\ \text{CONTRAINDICATED__IN},\ \text{MANAGES},\ \text{USED__WITH},\ \text{ASSOCIATED__WITH},\ \text{HAS__TARGET},\ \text{SPECIFIES__TARGET},\ \text{MEASURED__BY},\ \text{REQUIRES__MONITORING},\ \text{HAS__PROTOCOL},\ \text{DESCRIBES__PROTOCOL},\ \text{MENTIONED__IN},\ \text{EQUIVALENT__TO}\ \}$$

These eighteen relation types are derived from recurring relationships in the corpus together with the structural and provenance requirements of the graph. Clinical relation types reflect relationships stated in the guideline content, while structural and provenance relations encode document organisation and traceability. The EQUIVALENT_TO relation is different again: it is created during normalisation rather than extracted from text.

Relation types are retained only where they represent a distinction needed by the graph and could be populated reliably from the corpus. For example, FIRST_LINE_FOR and SECOND_LINE_FOR remain separate because merging them into RECOMMENDED_FOR would remove the treatment-sequence distinction expressed in the source guidelines.

Pharmacological relations

These relations record treatment decisions. The guidelines give a drug's position in a treatment sequence, not just an association with a condition, so the schema needs to represent sequence. "First-line" and "second-line" refer to the order treatments are tried in, not to drug quality.

FIRST_LINE_FOR and SECOND_LINE_FOR are kept as separate types because the guidelines themselves use the terms "first-line" and "second-line" directly in the recommendation text National Institute for Health and Care Excellence (2015b). RECOMMENDED_FOR is a weaker type for cases where a guideline endorses a treatment without placing it in that hierarchy, which shows up often in guidelines whose tiered-care structure does not map cleanly onto first- or second-line. CONTRAINDICATED_IN records that a drug should not be used in a given context; leaving it out would have meant the graph could only represent positive recommendations. MANAGES links an intervention to a condition without naming a treatment-sequence position, which is common for non-pharmacological interventions such as lifestyle changes. USED_WITH records two agents named together in one recommendation. NICE recommendation 1.13.1 National Institute for Health and Care

Excellence (2015b) names modified-release metformin and an SGLT-2 inhibitor in a single statement, and representing that as two separate FIRST_LINE_FOR edges would lose the fact that the source names them together. ASSOCIATED_WITH is used for a weaker descriptive relation where the source links a clinical concept to another concept without stating a treatment recommendation, contraindication, or management relation.

Three relation types were considered here and dropped. PREFERRED_OVER would assert a direct comparative ranking between two drugs. In the four guidelines examined, preference between drugs was stated inconsistently and with population-specific qualifications attached, which made reliable extraction impractical; the same preference order can already be inferred by comparing the FIRST_LINE_FOR and SECOND_LINE_FOR edges on a shared condition node, so a dedicated relation was not needed. DOSAGE_OF was dropped because dosage statements in the corpus are heavily conditioned on population, timing, renal function, age, and comorbidity, and a single binary edge cannot represent that. CAUSES, for adverse effects, was dropped because adverse-effect statements are written in prose too variable to decompose reliably into clean drug-effect pairs.

Monitoring and measurement relations

A condition may be linked to a glycaemic threshold through HAS_TARGET. This relation is created only when the surrounding source context identifies the diabetes type unambiguously. If the context does not distinguish Type 1 from Type 2 diabetes, the GlycaemicTarget remains in the graph with its provenance but no disease-specific HAS_TARGET relation is added. The target may be linked to the relevant test through MEASURED_BY, and a drug or condition may be linked to a monitoring protocol through REQUIRES_MONITORING.

For example, when a passage explicitly identifies Type 1 diabetes and states a numerical glycaemic target, the resulting target can be linked to the global Type 1 Diabetes condition node. A numerical target that does not establish whether the surrounding statement concerns Type 1 or Type 2 diabetes is retained without such a condition-specific edge. This avoids assigning a disease scope that is not supported by the local source context.

SPECIFIES_TARGET links a Recommendation or Table node to the GlycaemicTarget it states explicitly. This edge is created only when the numerical target is extracted from an element with a known identity in the graph; targets extracted from general prose chunks, which have no such element to point back to, receive no SPECIFIES_TARGET edge. HAS_PROTOCOL links a clinical concept to a MonitoringProtocol associated with its monitoring requirements. DESCRIBES_PROTOCOL links a source element, such as a recommendation or table, to the MonitoringProtocol whose procedural content it describes.

This chain exists because comparing thresholds across organisations is one of the central objectives of the graph, and that comparison needs a structured place to point to. HAS_TARGET points to a formally defined GlycaemicTarget node, which is what gets normalised and later carries an EQUIVALENT_TO edge. REQUIRES_MONITORING is kept separate from HAS_TARGET because a monitoring protocol is procedural, not numerical; it specifies what to check and how often, not a threshold value. MEASURED_BY

connects a GlycaemicTarget to the diagnostic test used to assess it, and scopes the unit-normalisation problem: two targets for the same condition, measured by the same test but expressed in different units, become candidates for an EQUIVALENT_TO edge once normalised. A FREQUENCY_OF relation was considered but dropped, since monitoring-frequency statements in the corpus sit inside conditional clauses that would need a much richer event model to represent properly.

Cross-guideline and provenance relations

These relations give the graph its multi-guideline and provenance structure. They connect organisations, documents, extracted content, and cross-guideline comparison results.

PUBLISHES, HAS_RECOMMENDATION, and CONTAINS_TABLE provide the main document-level provenance links. An Organisation connects to its Guideline nodes through PUBLISHES. A Guideline connects to its Recommendation nodes through HAS_RECOMMENDATION and to its Table nodes through CONTAINS_TABLE. Together, these relations preserve the connection between document content and the organisation that issued the guideline.

MENTIONED_IN is a provenance relation linking a global entity to the guideline in which it occurs. These links are created during normalisation from the entity's recorded guideline mentions and are retained independently of any more specific clinical relations involving the same entity.

EQUIVALENT_TO is not extracted from text. It is created during normalisation when two GlycaemicTarget nodes from different guidelines share the same metric, operator, and normalised value after unit conversion. It provides the mechanism used in this work to represent cross-guideline agreement on a numerical threshold.

Two further types were considered and dropped from this group. SUPERSEDES, for one recommendation replacing an older one from the same organisation, would need a model of guideline update cycles over time, which is out of scope for a corpus frozen at a single point in time. CONTRADICTS was also considered, but disagreement between organisations was not represented as an explicit relation in the source documents. Differences can instead be identified by comparing FIRST_LINE_FOR or HAS_TARGET edges associated with the same condition. Whether such a difference constitutes a clinical contradiction is not determined by the graph.

3.4.4. Schema Design Rationale

The schema balances two requirements that pull in opposite directions. It needs enough types to preserve distinctions that matter in the source guidelines, but every additional type also has to be populated reliably. For example, collapsing FIRST_LINE_FOR and SECOND_LINE_FOR into RECOMMENDED_FOR would make extraction simpler, but it would remove treatment order from the graph. At the other extreme, a relation such as DOSAGE_OF looks useful in principle but would need more structure than a single edge can carry. No systematic optimisation over alternative schema sizes is performed. The retained types are those for which the distinction is useful to the queries supported

by the graph and for which the corpus provides enough structure to populate the type consistently. The rejected relation types and their individual reasons are discussed with the corresponding relation groups above.

Figure 3.2 illustrates the node-scoping structure and representative relations in the Knowledge Graph. The node columns show representative node types from the global, structural, and organisation-scoped categories. The arrows show representative examples from the three relation groups: pharmacological, monitoring and measurement, and cross-guideline and provenance. The complete set of eighteen relation types is defined in $\mathcal{T}_E$ above.

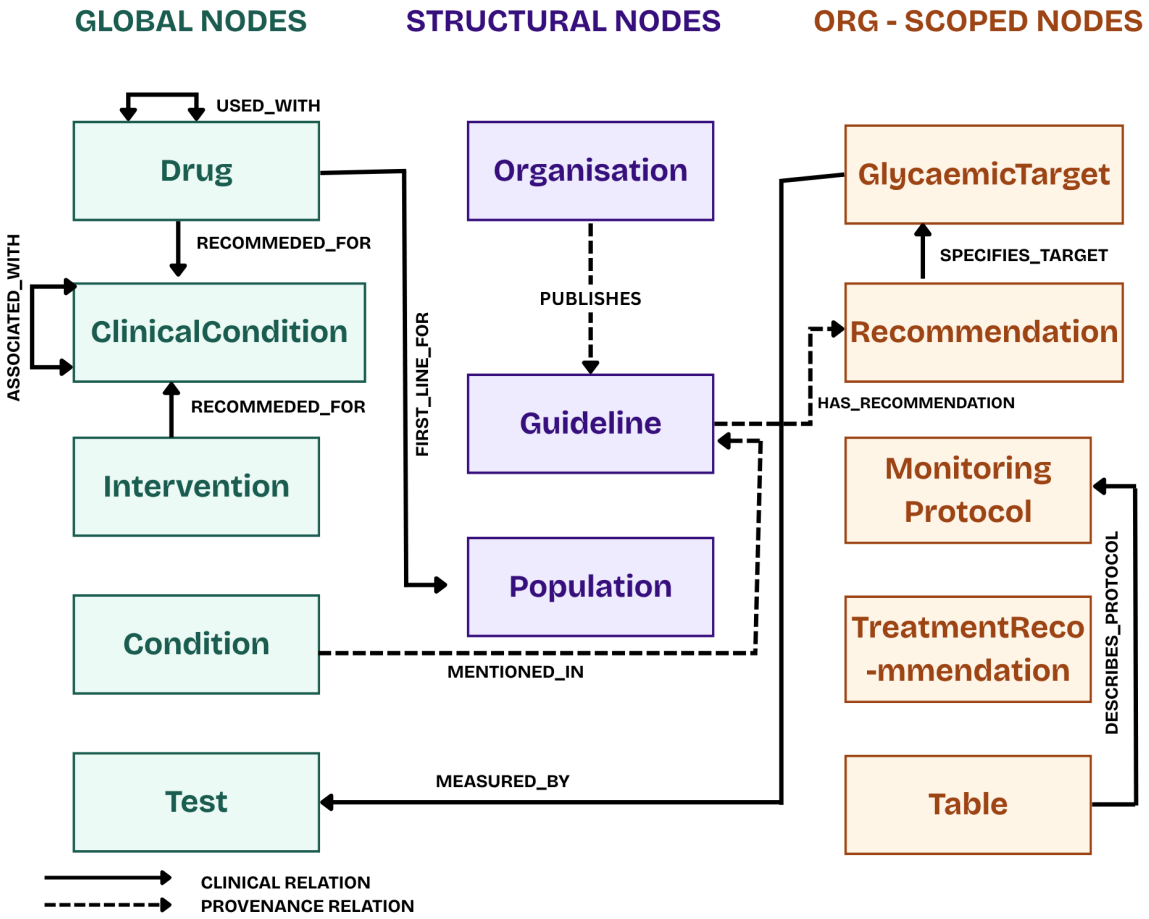


Figure 3.2.: Knowledge Graph Schema

Global nodes are what let edges cross between guidelines at all; leaving them out would mean each guideline produces its own isolated subgraph. Organisation-scoped nodes prevent the opposite problem. Leaving them out would merge every guideline's specific position into one shared node per concept, erasing the variation that cross-guideline comparison is meant to surface.

3.5. Entity and Relation Extraction

The extraction stage reads the structured records produced by φ and identifies the clinical entities and relations that become nodes and edges in the graph. This is where content elements from $\mathcal{T}_C$ turn into typed graph elements drawn from $\mathcal{T}_V$ and $\mathcal{T}_E$, and it is where the main architectural decisions sit. Three matter most. Which entity types go to which extraction layer, how the language model is constrained, and how identity is kept consistent across documents.

Two difficulties were particularly important in this corpus. The first is terminological precision. Clinical guidelines use controlled vocabulary where small differences in wording carry large differences in meaning; in the corpus documents, a drug described as "recommended" is not the same as one described as "preferred", which is not the same again as one "considered as an option". An extraction method that treats these as equivalent produces a graph that misstates how strongly a guideline actually recommends something. The second is cross-document consistency: the same concept can be worded differently across guidelines, for example "type 2 diabetes" in one and "non-insulin-dependent diabetes mellitus" in another, so extraction has to produce identifiers that match across documents, not only within one. Named entity recognition and relation extraction as techniques are background covered in Chapter 2 Nasar et al. (2021); Zhao et al. (2024); what this chapter contributes is not the techniques themselves but the specific architecture through which they are applied to a multi-guideline clinical corpus.

The Extraction Function

Extraction is usually split into entity recognition and relation extraction as separate steps Nasar et al. (2021); here both are combined into one function:

$$\xi : 2^{\mathcal{E}} \rightarrow \mathcal{V} \times \mathcal{R} \tag{3.2}$$

where $2^{\mathcal{E}}$ is the power set of element records produced by φ, $\mathcal{V}$ is the set of typed entity records, and $\mathcal{R}$ is the set of typed relation records. An entity record $v \in \mathcal{V}$ is a tuple

$$v = (\mathrm{id}_v,\ \tau_v,\ \mathrm{name}_v,\ \pi_v,\ m_v)$$

where $\tau_v \in \mathcal{T}_V$ is the entity type, $\mathrm{name}_v \in \mathbb{S}$ is the canonical surface form, π_v is the provenance inherited from the source element, and $m_v \in \{\textsc{Rule}, \mathrm{LLM}\,\emptyset\}$ records which layer produced it. A relation record $r \in \mathcal{R}$ is

$$r = (\mathrm{id}_r,\ \mathrm{src}_r,\ \psi_r,\ \mathrm{dst}_r,\ \pi_r,\ m_r)$$

where $\psi_r \in \mathcal{T}_E$ is the relation type, src_r and dst_r are entity identifiers, π_r is provenance, and m_r is the method label. This method label is what makes each layer's contribution checkable on its own during normalisation and later evaluation, instead of only as a combined result.

3.5.1. Rule-Based Extraction

Rule-based extraction applies pattern matching and structural heuristics directly to the element records to pull out a specific, limited set of entity and relation types. Rather

than trying to extract everything, the rule layer only targets types that have a reliable structural or lexical signal in the guideline text.

A type is assigned to the rule layer when its instances have a predictable syntactic or structural form that is sufficiently consistent in the document families where that type occurs. Types that do not meet this condition are assigned to the contextual extraction layer.

GlycaemicTarget entities A numerical target has a local structure that can be decomposed directly. For example,

$$\text{FPG} \geq 7.0 \text{ mmol/L}$$

contains the metric `FPG`, the operator $\geq$, the value `7.0`, and the unit `mmol/L`. These four fields are captured separately by the rule layer: a regular expression matches the metric name and unit as fixed anchors, with the operator and numeric value extracted as separate groups in between. The gap between the metric name and the operator is allowed to contain up to 40 or 60 characters of intervening text, depending on the metric, so the same pattern still matches when a guideline states the target inside a longer sentence rather than as a short standalone expression. A second instance illustrates this with a different metric and unit,

$$\text{HbA1c} \geq 48 \text{ mmol/mol},$$

extracted by a separate but structurally identical pattern. The surrounding sentence may vary, but the target itself retains the same four-field structure in both cases.

MonitoringProtocol entities come from two sources. Prose containing continuous glucose monitoring or time-in-range terminology, matched by terms such as *time in range*, *TIR*, and *CGM*, produces one MonitoringProtocol node per organisation. These mentions can recur throughout a document, so creating a new node for every occurrence would represent the same monitoring concept several times. Table captions provide the second route. A caption containing monitoring-related terms such as *monitor*, *screen*, *follow-up*, or *frequency* produces a MonitoringProtocol associated with that table. Both cases have an explicit lexical or structural signal and are therefore handled by the rule layer without a language-model call.

TreatmentRecommendation entities are extracted from tables rather than from numbered Recommendation elements. A table whose caption contains treatment-related vocabulary such as *treat*, *therap*, *pharmac*, *drug*, *medic*, or *insulin* produces a TreatmentRecommendation node. This is distinct from a Recommendation node, whose boundary and recommendation number come from the numbered guideline statements identified during document structuring. The schema keeps the two types separate because they represent two different source structures: one is an explicitly numbered guideline statement and the other is a treatment summary identified from a table.

Canonical anchor nodes Condition and Test do not follow the same extraction route as the entity types above. A fixed list of five Condition nodes (Type 1 Diabetes, Type 2

Diabetes, Prediabetes, Intermediate Hyperglycaemia, and Gestational Diabetes) is created before extraction begins rather than being extracted independently from each document. TEST nodes are created when extraction encounters a named test, such as HbA1c or CGM, but repeated occurrences of the same test name resolve to the same global node. The identity problem is therefore different in the two cases. For CONDITION, identity is fixed before extraction. For TEST, identity is resolved by exact name matching when a test is encountered. Both types can subsequently be supplied to the LLM layer as known entities, as described in Section 3.5.2.

Rule-based extraction is interpretable and deterministic. Every entity traces back to the pattern that produced it, and a false positive can be investigated by examining or modifying that pattern, which is one reason rule-based systems remain widely used in practice Chiticariu et al. (2013). Its limitation is coverage. In the guidelines processed here, drug indications stated across several clauses, contraindications buried in qualifying remarks, and population-specific modifications inside conditional phrasing eluded reliable pattern matching. This coverage gap motivates the LLM-based layer, since language models can extract structured information from text with little task-specific training data Xu et al. (2024); Caufield et al. (2024).

3.5.2. LLM-Based Extraction

The contextual extraction layer identifies entities and relations that need contextual reading, where pattern matching alone won't work, and targets the entity types that are beyond what a rule can reliably reach. Both layers operate on the structured document records produced by φ, but the contextual layer additionally receives page-level facts already identified by the rule layer as grounding context. The two layers are therefore separated by target category rather than by independent execution.

The constrained generation context. The main risk in applying a language model to clinical text is hallucination Huang et al. (2025). An incorrect drug indication or a fabricated contraindication is a worse failure here than in most extraction tasks, so the design leans toward traceability and grounding over raw coverage. Prior work on schema-constrained prompting shows that giving a model an explicit output schema as a generation constraint reduces hallucination in knowledge-base population; the LLM layer here follows the same principle, operating under a constrained generation context

$$\mathcal{C} = (\mathcal{T}_V,\ \mathcal{T}_E,\ \mathcal{K})$$

where $\mathcal{T}_V$ and $\mathcal{T}_E$ are the allowed entity and relation types, and $\mathcal{K}$ is the page-level set of structured facts already identified by deterministic extraction. In the implementation, this context contains numerical targets and recommendation records associated with the current page. Page-level scope is used for $\mathcal{K}$ for two reasons. A chunk-level scope is too narrow because a numerical target or recommendation identified in one element on a page may provide relevant grounding for another element on the same page. A document-level scope is too broad, since passing facts from the entire guideline would add information unrelated to the passage being processed. Page-level scope therefore preserves nearby grounding information while keeping the context focused on the current page.

$\mathcal{C}$ is injected into the model's prompt as a prompt-level constraint. The system prompt lists the allowed entity and relation types directly and instructs the model to restrict its output to types already present in $\mathcal{T}_V$ and $\mathcal{T}_E$. This is an instruction rather than an enforced constraint. The API call guarantees only that the response parses as structured output. The parser does not reject records whose type falls outside the declared sets, so compliance rests on the model following the instruction. In the reported run, two relation records used undeclared relation types (researching and intervention). These records were excluded from the final graph during integration.

Together, these two constraints define what the model is asked to do. It classifies and structures content from the input passage, using surface forms and identities the rule layer and schema have already set. Nothing beyond the passage, the schema, and the page-level rule-derived context K is supplied to it. Whether the model also draws on its own parameters is not something the output can tell us.

LLM extraction targets. Four entity types go to the LLM layer because rule-based methods cannot reliably reach them. DRUG entities appear throughout guideline prose in varying syntactic positions — as the subject of a recommendation clause, the object of a contraindication statement, or a member of an enumerated drug class — and picking them out needs the kind of contextual reading a language model does well. Metformin, for example, may appear as the treatment being recommended, as part of a combination, or inside a statement that qualifies when it should be used. Its syntactic position therefore does not provide the stable cue available for a numerical target. POPULATION entities, describing the patient group a recommendation applies to, are often long noun phrases with several qualifying modifiers, which makes pattern matching unreliable. CLINICAL-CONDITION entities, covering comorbidities and complications that interact with diabetes management, are worded very differently across the four guideline families. INTERVENTION entities, covering non-pharmacological approaches such as lifestyle change or surgery, use terminology that varies enough between organisations that a fixed dictionary lookup would not hold up.

For relations, the LLM layer targets ones that need reasoning across more than one sentence: CONTRAINDICATED_IN relations stated across conditional clauses, USED_WITH relations implied by a description of combination therapy, and MANAGES relations asserted through treatment narrative, not an explicit recommendation statement.

Separation and auditability. Figure 3.3 makes one design decision explicit. The rule and contextual components are assigned disjoint entity categories rather than competing to extract the same types. Contextual extraction can use facts already identified by the rule layer as grounding information, but it does not act as a fallback that simply fills every gap left by the rules. Method labels retained on extracted records allow the contribution of each extraction route to be inspected separately.

The output of both layers, the entity records in $\mathcal{V}$ and relation records in $\mathcal{R}$, passes to the normalisation stage, which resolves duplicates, converts units, and integrates both record sets into the finished graph G, described next in Section 3.6.

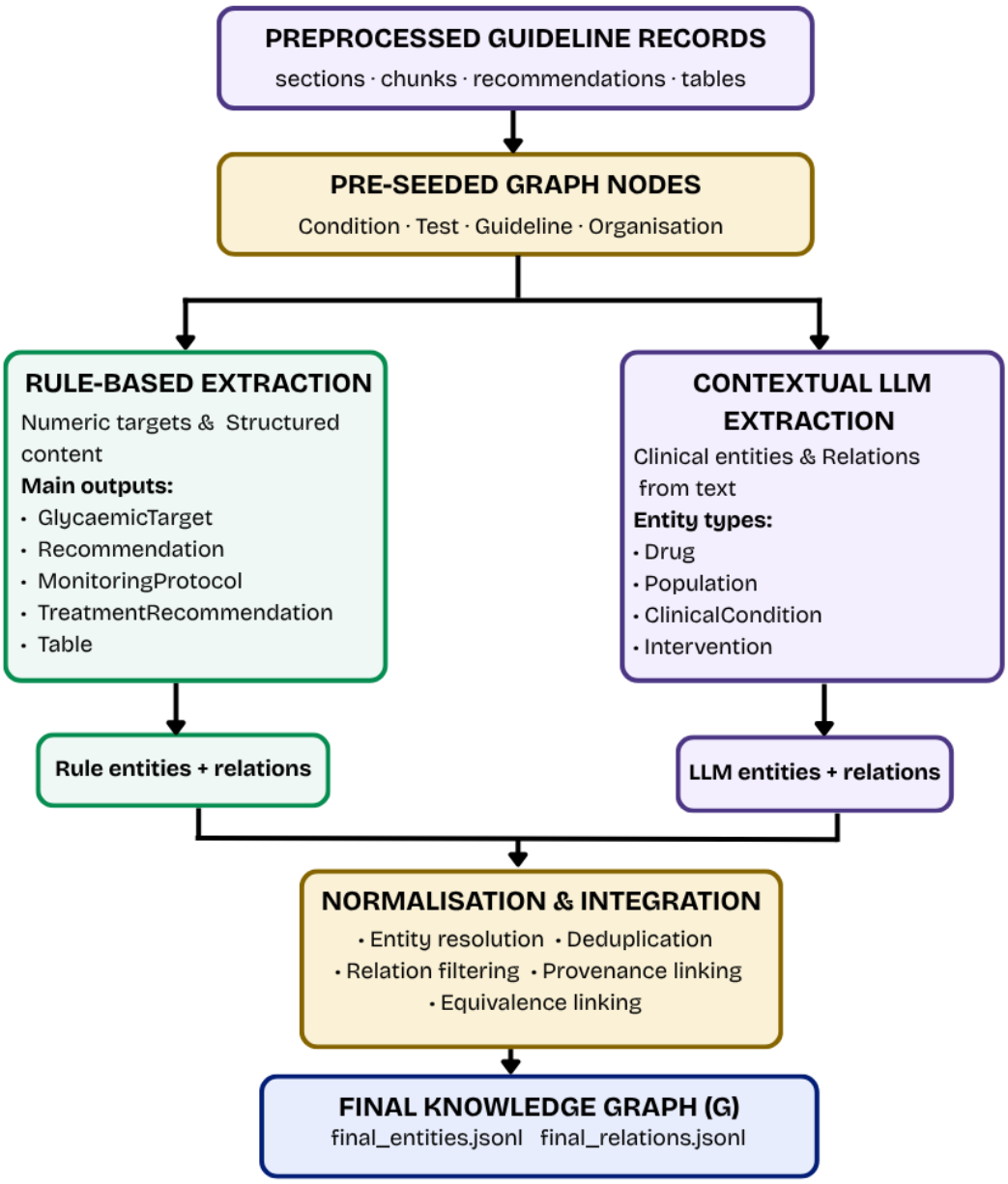


Figure 3.3.: Layered extraction architecture

3.6. Normalisation and Graph Integration

The normalisation stage takes the entity records $\mathcal{V}$ and relation records $\mathcal{R}$ produced by ξ and integrates them into the unified graph G. It addresses three problems the extraction stage necessarily leaves unresolved — entity duplication, terminological variation, and unit inconsistency. Entity deduplication is studied in the entity resolution and record linkage literature Getoor and Machanavajjhala (2012); Christen (2012); Brizan and Tansel (2015), and mapping biomedical mentions to standard concepts is a long-standing task in clinical NLP, for example for disease names Singh et al. (2013). Unit normalisation is needed here

because clinical measurements are reported on different scales across guidelines Hoelzel et al. (2004). The contribution of this stage is applying both steps within one multi-guideline pipeline.

The normalisation stage is formalised as a function

$$\eta : (\mathcal{V} \times \mathcal{R}) \to G \tag{3.3}$$

where $G = (V, E, \mathcal{T}_V, \mathcal{T}_E, \phi, \psi, \lambda)$ is the graph defined in Section 3.4.1. η has to satisfy two properties. *Provenance monotonicity*: merging two records never reduces the provenance carried by the result, since the provenance of the merged element is the union of everything that went into it.

3.6.1. Entity Deduplication

Both extraction layers can independently identify the same real-world entity from different passages of the same document, or from different documents entirely, producing more than one record for the same concept. Loading those duplicates into the graph without resolving them first would leave several near-identical nodes standing in for one entity, splitting relations that should converge on a single node across all of them and making cross-guideline queries incomplete.

Deduplication uses a type-specific equivalence relation $\sim_\tau$ over entity records. Two records e_i, e_j of type τ count as duplicates exactly when $e_i \sim_\tau e_j$, and are then merged into one node whose provenance is the union of π_{e_i} and π_{e_j}. This relation takes two different forms, depending on the structural properties of the type.

Name-bearing entity types. For types whose identity is a canonical name string — Drug, Condition, ClinicalCondition, Intervention, and Test — the equivalence relation is

$$e_i \sim_\tau e_j \iff \sigma(\text{name}(e_i)) = \sigma(\text{name}(e_j))$$

where $\sigma : \mathbb{S} \to \mathbb{S}$ normalises a surface form in three steps: it collapses whitespace and strips leading and trailing spaces, applies Unicode case folding, and then maps the result through a type-specific canonicalisation dictionary $\Delta_\tau : \mathbb{S} \to \mathbb{S}$ that resolves known surface-form variants to one canonical representative. In the implementation, separate alias mappings are used for different name-bearing types, including tests, drugs, and clinical conditions. For example, the test aliases `fpg`, `fasting plasma glucose`, and `fasting glucose` all resolve to `FPG`. These dictionaries are fixed at schema design time and not updated while processing runs; any surface form not already in Δ_τ is treated as a new entity, not a variant of an existing one. Because everything here is deterministic, $\sim_\tau$ is exactly reproducible — the same input records and the same dictionary always produce the same deduplication output.

No similarity threshold or fuzzy string matching is applied in this deduplication step. Deterministic matching is chosen over probabilistic matching because a fuzzy match that incorrectly merges two distinct drug or condition names would put an error into the graph that is hard to catch by inspection afterward. The trade-off is reduced recall, discussed in Chapter 4.

Structured numerical entity types. For types whose identity comes from a combination of numerical and categorical fields, not a name string — GlycaemicTarget and MonitoringProtocol — the equivalence relation is defined over a composite key,

$$e_i \sim_\tau e_j \iff \kappa_\tau(e_i) = \kappa_\tau(e_j)$$

where $\kappa_\tau : \mathcal{E} \to \mathcal{K}_\tau$ projects a record onto its defining fields. For GlycaemicTarget, the key is

$$\kappa_{\mathrm{GT}}(e) = (\mathrm{metric}(e),\ \mathrm{units}(e),\ \mathrm{op}(e),\ \mathrm{value}(e))$$

Two records are duplicates only if all four fields match exactly. As with the name-bearing types, no tolerance is applied anywhere in this comparison.

Scoping of deduplication. For global node types, the relation applies across the whole corpus. Two records from different guidelines that satisfy $\sim_\tau$ merge into one global node. For organisation-scoped types, deduplication stays within a single issuing organisation. Two GlycaemicTarget records from different organisations that happen to satisfy $\kappa_{\mathrm{GT}}(e_i) = \kappa_{\mathrm{GT}}(e_j)$ are not merged, because they represent two independent clinical positions that happen to agree numerically, not one shared statement. That agreement is instead expressed through an EQUIVALENT_TO edge, created during unit normalisation, described next.

3.6.2. Unit Normalisation

As established in Section 3.2.1, different organisations express the same clinical quantities in different units. Comparing numerical thresholds across guidelines only means anything once every value is expressed in one common unit system.

Clinical quantities such as HbA1c are reported on different scales across guideline families, for example NGSP percent and IFCC mmol/mol, which are related by a linear regression Hoelzel et al. (2004), so thresholds from different sources have to be converted before they can be compared. This is made easier by never merging organisation-scoped nodes during deduplication, which keeps each guideline's own threshold as a distinct node that can be compared against another's.

Unit normalisation is applied to GlycaemicTarget entities during integration. Plasma glucose values given in milligrams per decilitre convert to millimoles per litre by

$$\mathrm{value}_{\mathrm{mmol/L}} = \frac{\mathrm{value}_{\mathrm{mg/dL}}}{18.0}$$

where the divisor is the standard factor for converting plasma glucose between the two units. For example, a stored value of 126 mg/dL becomes 126/18.0 = 7.0 mmol/L. A target written as 126 mg/dL can therefore be compared directly with a target written as 7.0 mmol/L once both are represented through the normalised field. This factor is not asserted independently of the corpus; several guidelines state both units for the same threshold directly, for example the IDF criteria Federation (2025) give FPG <6.1 mmol/L (110 mg/dL) for normal fasting glucose and FPG ≥7.0 mmol/L (126 mg/dL) for the diabetes diagnostic threshold, and a 2-hour post-load value of <10 mmol/L (180 mg/dL).

Dividing each mg/dL figure by 18.016 reproduces the mmol/L figure given alongside it, confirming the pipeline's rounded divisor of 18.0 against the corpus's own stated equivalences rather than an external constant. The pipeline uses 18.0 rather than 18.016; for the threshold values present in this corpus the two give identical results once rounded to one decimal place. The original value and unit are kept as properties on the node alongside the normalised one, so the source document's original expression is never lost. Glycated haemoglobin appears in the corpus both as a percentage and in IFCC units (mmol/mol); both forms are retained, and each is converted to the other using the linear relation from the IFCC reference method Hoelzel et al. (2004):

$$\text{value}_{\text{mmol/mol}} = \frac{\text{value}_{\%} - 2.152}{0.09148}$$

As with plasma glucose, the source value and unit stay on the node alongside the converted one.

After unit normalisation, the implementation rounds the normalised plasma-glucose value to one decimal place before comparison. Two GlycaemicTarget nodes from different guidelines are considered equivalent if

$$\begin{aligned} e_i \equiv e_j \iff & \text{metric}(e_i) = \text{metric}(e_j) \ \wedge \ \text{op}(e_i) = \text{op}(e_j) \ \wedge \\ & \text{round}_1\Big(\text{value}_{\text{mmol/L}}(e_i)\Big) = \text{round}_1\Big(\text{value}_{\text{mmol/L}}(e_j)\Big) \end{aligned}$$

This requires exact equality of the rounded value, with no additional tolerance band. This is workable for the thresholds compared in this corpus because values such as the fasting plasma glucose threshold of 7.0 mmol/L resolve to the same rounded value after normalisation. For each pair (e_i, e_j) satisfying these conditions and coming from different organisations, a directed EQUIVALENT_TO edge is created between them.

Conceptually, equivalence detection belongs to normalisation because it depends on normalised target values. In the implementation, the normalised plasma-glucose field and the corresponding EQUIVALENT_TO relations are computed while assembling the rule-derived target records, before the later global deduplication pass. This step is where the fourth contribution of the chapter is actually realised. No guideline states outright that its own threshold agrees with another organisation's; the relation is not extracted from text at all. It exists only because unit conversion makes independently stated thresholds directly comparable once they are expressed the same way. Normalisation is therefore not only a cleanup step; it is also the point in the pipeline that produces a class of cross-guideline relation that no single-guideline pipeline, and no extraction method working on raw text, could produce on its own.

3.6.3. Layered Output Integration

Merging records that refer to the same concept onto one canonical identifier follows the same principle used by controlled vocabulary resources in biomedicine Bodenreider (2004).

The final step merges the deduplicated outputs of the rule-based and LLM-based layers into one set of nodes and edges. When the LLM layer produces an entity record, the entity-resolution step checks it against the index of canonical entities already established

by the rule layer and the pre-seeded anchor nodes. If it finds a match under $\sim_\tau$, the LLM record is discarded and the canonical entity is referenced instead; only when no canonical match exists does the record keep an `llm_` identifier prefix and get retained as a new entity. This applies the entity-resolution rule of $\mathcal{C}$ again at the integration stage, in addition to the prompt-level instruction already described in Section 3.5.2.

Relations that resolve to the same source node, relation type, and destination node are deduplicated during integration; the implementation retains one relation record after rewiring rather than creating parallel duplicate edges. Relations present in only one layer's output are kept as they are. The two layers are meant to complement, not duplicate, each other, so a relation found only by the rule layer, or only by the LLM layer, is not treated as less valid for that reason.

The integrated, normalised, and deduplicated set of nodes and edges is then loaded into the graph database, producing the finished graph $G = (V, E, \mathcal{T}_V, \mathcal{T}_E, \phi, \psi, \lambda)$ as defined in Section 3.4.1. Where an entity or relation is produced by an extraction component, its extraction-method metadata is retained so that rule-derived and language-model-derived records can be distinguished during analysis.

3.7. Question Generation from the Knowledge Graph

The final stage of the pipeline uses the finished graph G as a controlled knowledge source for generating clinical questions. Two properties are both wanted here and are hard to get at the same time. Linguistic variety means that questions need not all use one fixed surface template for a given relation type. Factual grounding means every claim made in a question, or embedded in its answer, has to trace back to a specific assertion already in the graph.

A purely template-based approach, substituting graph values into fixed sentence patterns, guarantees grounding but at a real cost to variety. Every question on the same relation type sounds the same, which is likely to make the catalogue predictable for a learner. Low diversity is also a recognised problem in knowledge-base question generation more broadly Bi et al. (2020). A purely generative approach, where a language model is simply prompted to write questions about diabetes, gets the variety but brings back the hallucination risk this chapter has already discussed. The model can produce a question whose factual claim is not actually supported by the corpus, or attribute a recommendation to the wrong organisation Huang et al. (2025). Language models can be prompted with structured knowledge-base input to generate questions Liang et al. (2023), and the hybrid design in this section builds on that idea while splitting the two responsibilities between the component suited to each. The graph decides what is asked, and the language model decides how it is phrased.

The Question Generation Function

Question generation is formalised as a function

$$\gamma : G \to \mathcal{Q} \tag{3.4}$$

mapping the graph to a catalogue of validated questions $\mathcal{Q}$. γ is built from three sub-functions applied in order — targeting, formulation, and validation — with the anchor produced by targeting carried forward through formulation so that validation can compare the final question against it. The factual content of every question is fixed by the first sub-function, before the language model is ever called, and checked again by the third, after the model has produced its output. The model itself, in the middle step, has no access to the full graph. It receives only the selected anchor as factual input and is instructed not to introduce information beyond it.

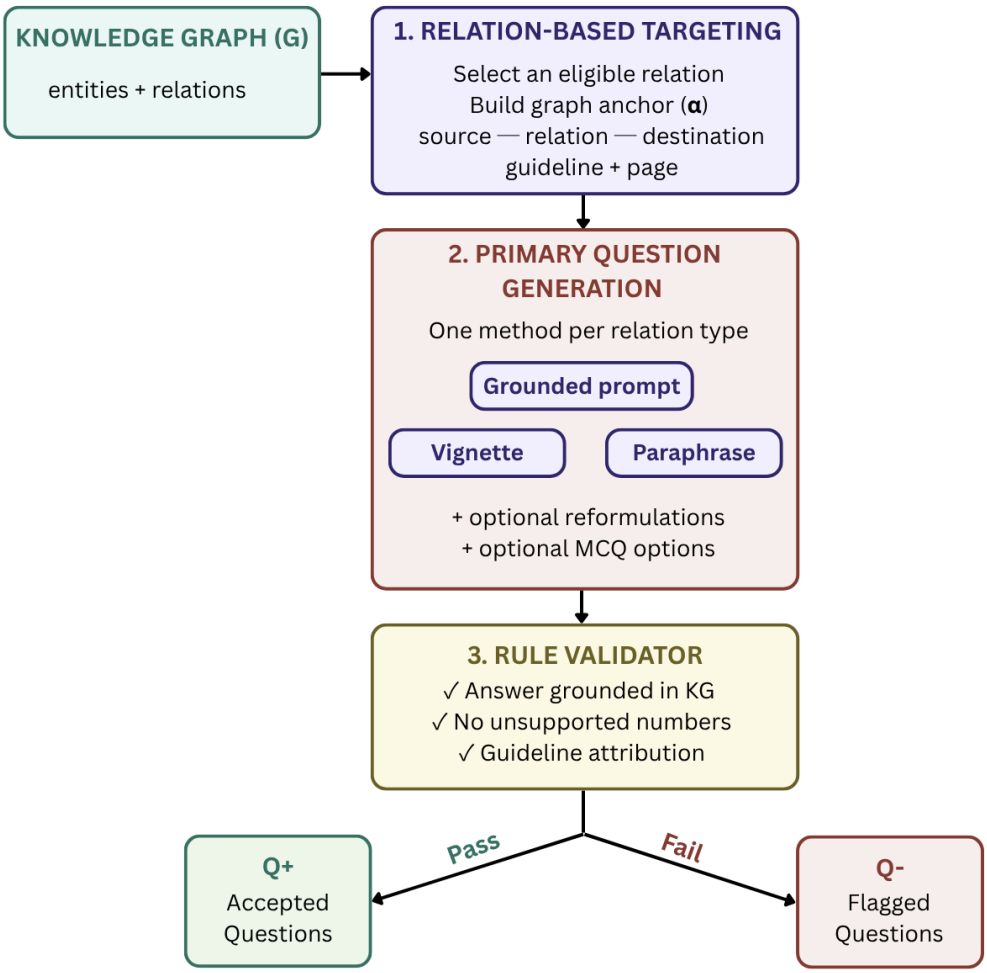


Figure 3.4.: Hybrid question generation pipeline.

3.7.1. Graph-Controlled Question Targeting

Generating questions from structured knowledge, not free text, is an established approach in question generation research Guo et al. (2024); Bi et al. (2020).

The first sub-function does targeting entirely by rule-based traversal of the graph, with no language model involved.

$$\alpha : G \to \mathcal{A} \times \mathcal{D}_\gamma$$

For each edge selected, α produces two things. A graph anchor $a \in \mathcal{A}$ is a frozen record of the fact the question will test, carrying the source node, relation type, destination node, guideline, and page. For each selected edge, the implementation determines a question type and one primary generation method from fixed relation-based mappings. MCQ eligibility is also predefined for selected relation types, while the primary difficulty label is assigned according to the generation method

Question generation is restricted to a selected subset of graph relation types for which a meaningful question form is defined. The implementation maps each eligible relation type to a question type and generation method; for example, FIRST_LINE_FOR maps to a *which* question, HAS_TARGET to a *what* question, CONTRAINDICATED_IN to a *yes/no* question, and MANAGES to a scenario-based *how* question. EQUIVALENT_TO edges are treated separately. They connect the same clinical concept across two guidelines, so questions derived from them test recall of one guideline's position and cross-guideline awareness at the same time.

The question taxonomy $\mathcal{T}_Q = \{\text{WHICH}, \text{WHAT}, \text{YES_NO}, \text{HOW}\}$ is decided entirely by the relation type of the anchor edge, not by the model. The mapping is a fixed lookup table set before any generation begins. FIRST_LINE_FOR edges produce *which* questions, HAS_TARGET edges produce *what* questions, CONTRAINDICATED_IN edges produce *yes/no* questions, and MANAGES edges produce scenario-based *how* questions. The distribution of question types across the catalogue is a direct function of the graph's edge distribution, not of any generative choice made downstream.

3.7.2. Language Model Question Formulation

The second sub-function uses a language model to produce the natural-language form of the question.

$$\beta : \mathcal{A} \times \mathcal{D}_\gamma \to \mathcal{Q}_{\text{raw}}$$

The model receives the anchor a as its only factual input, given as a frozen context block, configured according to δ. It is instructed not to introduce drug names, numerical values, targets, or guideline references that are absent from the anchor. In vignette-style formulations it may additionally describe the patient and symptoms in words; such details are not drawn from the anchor, and the validator checks them only for numerical values. Its intended role is therefore linguistic transformation: it may vary sentence structure, framing, and formality while preserving the factual assertion represented by the anchor.

This is a tighter version of the same constraint used during extraction. In extraction, the model was confined to a predefined schema of types and canonical names. Here, it is confined further, to the single factual assertion in one anchor. The same principle runs through both stages. Its intended role is to classify or rephrase information supplied through the structured input rather than to introduce additional clinical content.

For each anchor, the method assigned to its relation type determines how the question is generated. The three generation methods are a graph-grounded prompt, a clinical

vignette, and a template paraphrase. For the grounded-prompt method, the model generates one primary question, and difficulty variants may be stored alongside it as reformulations within the same record. Each selected anchor produces at most one primary question record in the catalogue, and exact duplicate question texts are skipped. For relation types that support it, the model also generates multiple-choice distractors. The correct answer is fixed by the anchor and is never generated. For eligible multiple-choice questions, the model is given a candidate pool sampled from real DRUG nodes in the graph and instructed to choose plausible distractors from that pool. This differs from distractor generation from free text, as in Gao et al. (2019), where distractors are generated from the passage rather than drawn from a set of known entities. The returned strings are not programmatically checked against the pool.

3.7.3. Rule-Based Validation

Language models can produce text that reads fluently but is not actually supported by their input Huang et al. (2025), so every generated question is checked against its anchor before it enters the catalogue.

$$\nu : \mathcal{Q}_{\text{raw}} \times \mathcal{A} \to \{0, 1\} \times \mathcal{Q}$$

The validator applies three checks, each comparing the generated question text against the frozen anchor.

The first check is answer-key consistency. For questions whose answer is an entity, the validator confirms that the answer named in the anchor is actually a node in the graph. For questions derived from HAS_TARGET edges, where the answer is a numerical threshold and not an entity name, it confirms that the number in the answer matches the property values of the destination node v_d in the anchor.

The second check is hallucinated number detection. Any number in the generated question that does not appear in the anchor is flagged as a possible hallucination. This matters most for scenario-based questions, where the model can be tempted to invent realistic-sounding patient details, ages, lab values, or disease duration, that are not grounded in the anchor at all.

The third check detects conflicting guideline attribution. If the relation is associated with one of the recognised organisations, the validator rejects a question that explicitly names a different organisation. A question that credits a recommendation to the wrong organisation is factually wrong even if the recommendation content itself is stated correctly.

Questions passing all three checks go into the validated catalogue $\mathcal{Q}^+$, with anchor, generation configuration, difficulty level, and validation status recorded as metadata. Questions failing one or more checks go into a separate flagged catalogue $\mathcal{Q}^-$ for manual review. They are not discarded, since a failed check can just as easily mean the rule was too conservative as mean there is a genuine factual error. This trades some precision for recall, at the cost of needing human review on the borderline cases, which is a reasonable trade-off given that a wrong clinical claim in the catalogue actually matters. The validator checks whether the fluency and variety introduced by the language model in β remain consistent with the content represented by the anchor in α. What the validator checks is consistency with the anchor, not clinical correctness. A question built on an incorrect graph

edge will pass all three checks, since the checks compare the question against the anchor, not against the source guideline. Chapter 4 reports how often this occurred in practice.

Worked example. Consider an anchor in which a CONDITION node is linked by HAS_TARGET to a GLYCAEMICTARGET representing FPG ≥ 7.0 mmol/L. The relation also retains the organisation and page metadata associated with the extracted assertion where available. The targeting function α selects this relation and freezes its source node, relation type, destination node, and provenance as the anchor. The formulation function β receives only this anchor and turns it into a question such as "What fasting plasma glucose threshold does this guideline specify?" The answer remains 7.0 mmol/L because it comes from the destination node, not from the language model. The validator ν then checks the generated question and answer against the same anchor. If the question instead introduced a threshold of 8.0 mmol/L, the numerical check would flag it.

Figure 3.4 shows how the design keeps these two responsibilities separate. The left-hand annotations show the fixed mapping from relation type to question type, decided entirely by the rule engine with no language model involvement. The right-hand annotations show the range of linguistically distinct question variants the model generates from a single anchor, none of which may go beyond what the anchor specifies. The validator at the foot of the pipeline checks the generated output against the original anchor using the three rules described above.

The composition is expressed as $(a, \delta) = \alpha(G)$, $q_{\text{raw}} = \beta(a, \delta)$, and $q = \nu(q_{\text{raw}}, a)$, making explicit that the validator ν consumes both the formulated question and the original anchor. This composed pipeline is the third contribution of this chapter. It represents question generation anchored to the graph while keeping the factual content fixed by the selected anchor. The factual content of each question comes from the graph in α. The language model in β turns this content into natural language, and the validator in ν checks the generated question against the original graph assertion. The answer-bearing relation therefore remains fixed by the anchor throughout generation. The language model contributes the wording of the question and, where applicable, the distractor options selected from the permitted candidate pool. This separation allows the pipeline to produce questions with varied wording while keeping them tied to a specific graph assertion. The concrete results of applying γ to the graph, including question counts, type distributions, and validation outcomes, are reported in Chapter 4.

4. Experimental Evaluation

This chapter instantiates the framework defined in Chapter 3 on a corpus of Type 1 and Type 2 diabetes guidelines. Chapter 3 defined a four-stage architecture together with the schema, extraction rules, and normalisation procedures configured for the diabetes guideline corpus; this chapter is the diabetes-specific instantiation and owns all concrete details: corpus composition, tool choices, implementation decisions, quantitative outcomes, and an analysis of where the pipeline succeeded and where it did not. The sections follow the pipeline stages: corpus selection (Section 4.1), document ingestion and preprocessing (Section 4.2), knowledge graph construction (Section 4.3), question generation (Section 4.4), evaluation of the question catalogue (Section 4.5), and limitations (Section 4.6).

4.1. Data Sources and Corpus

The corpus is drawn from four guideline-producing organisations: the American Diabetes Association (ADA)[1], the National Institute for Health and Care Excellence (NICE)[2], the International Diabetes Federation (IDF)[3], and Diabetes Canada[4]. Together they satisfy the four selection criteria defined in Section 3.2. The corpus comprises eight documents, each processed as one ingestion unit. Table 4.1 reports the structural properties measured at the output of the ingestion stage. The IDF document served as the initial development document for the extraction pipeline before the implementation was extended to the other three guideline families.

4.1.1. Corpus against Selection Criteria

Geographic and systemic diversity. The four organisations publish from different national and international contexts: the ADA in the United States, NICE in England and Wales, Diabetes Canada in Canada, and the IDF as an international federation. Selecting organisations from more than one context gives variation in guideline origin, publishing convention, and recommendation structure, rather than restricting the corpus to a single publisher.

Clinical coverage. Across the eight documents the corpus addresses all five clinical domains identified in Section 3.2: diagnosis and classification (ADA Table 2.1, Canada

[1] https://www.diabetes.org/
[2] https://www.nice.org.uk/
[3] https://www.idf.org/
[4] https://www.diabetes.ca/

Organisation	Document	Year	Pp.	Rec.	Tab.	Units
ADA	Standards of Care in Diabetes	2026	377	176	113	mg/dL
NICE	NG17: Type 1 Diabetes in Adults	2026	64	100	0	mmol/L, mmol/mol
NICE	NG28: Type 2 Diabetes in Adults	2026	124	116	0	mmol/L, mmol/mol
IDF	Global Clinical Practice Recommendations	2025	112	10	21	mmol/L
Diabetes Canada	CPG 2018, Chapter 3	2018	6	2	5	mmol/L
Diabetes Canada	CPG 2018, Chapter 4	2018	4	2	1	mmol/L
Diabetes Canada	CPG 2018, Chapter 8	2018	5	5	1	mmol/L
Diabetes Canada	CPG 2018, Chapter 9	2018	7	8	1	mmol/L
Total	*8 documents, 4 organisations*		**699**	**419**	**142**	

Table 4.1.: Corpus documents and structural properties.

Ch. 3), screening (NICE NG17 Section 1.1, Canada Ch. 3), glycaemic targets (all four organisations), pharmacological management (ADA Section 9, NICE NG28, IDF, Canada Ch. 8–9), and monitoring (NICE NG17 Section 1.5, Canada Ch. 4). No domain is covered by a single organisation only.

Format diversity. NICE documents present recommendations as numbered statements, and convey their strength through wording rather than a grade label. In the versions processed here, the NICE guidelines contained [N] embedded tables. The ADA document combines numbered recommendations, each with an evidence grade, with 113 embedded tables in a three-column page layout. The IDF uses prose and two-column care-tier tables with no structured recommendation numbering. The Canada chapters are journal articles with graded recommendations embedded in continuous clinical commentary.

Recommendation specificity. All documents contain actionable recommendations specifying drug choices, numerical targets, or monitoring schedules. High-level policy documents were excluded because they do not produce meaningful pharmacological or measurement entities in the schema.

4.1.2. Structural Heterogeneity in Practice

NICE contributes 496 chunks, 216 structured recommendations across its two documents and zero tables; the ADA contributes 1,832 prose chunks, 176 structured recommendation elements, and 113 reconstructed tables; the IDF contributes 10 recommendation elements and 21 tables; the Canada chapters contributes 107 chunks, 8 tables and 17 recommendations. The measurement unit column illustrates the unit inconsistency: the ADA uses mg/dL for plasma glucose while the other three use mmol/L, and NICE reports HbA1c in

mmol/mol rather than as a percentage. These three dimensions of heterogeneity directly shaped every design decision in the ingestion and extraction stages.

Figure 4.1 shows the number of chunks, recommendations, tables, and sections extracted per organisation, visualising the same structural volume reported in Table 4.3 and highlighting NICE's absence of tables and the ADA's dominance in chunk and table count..

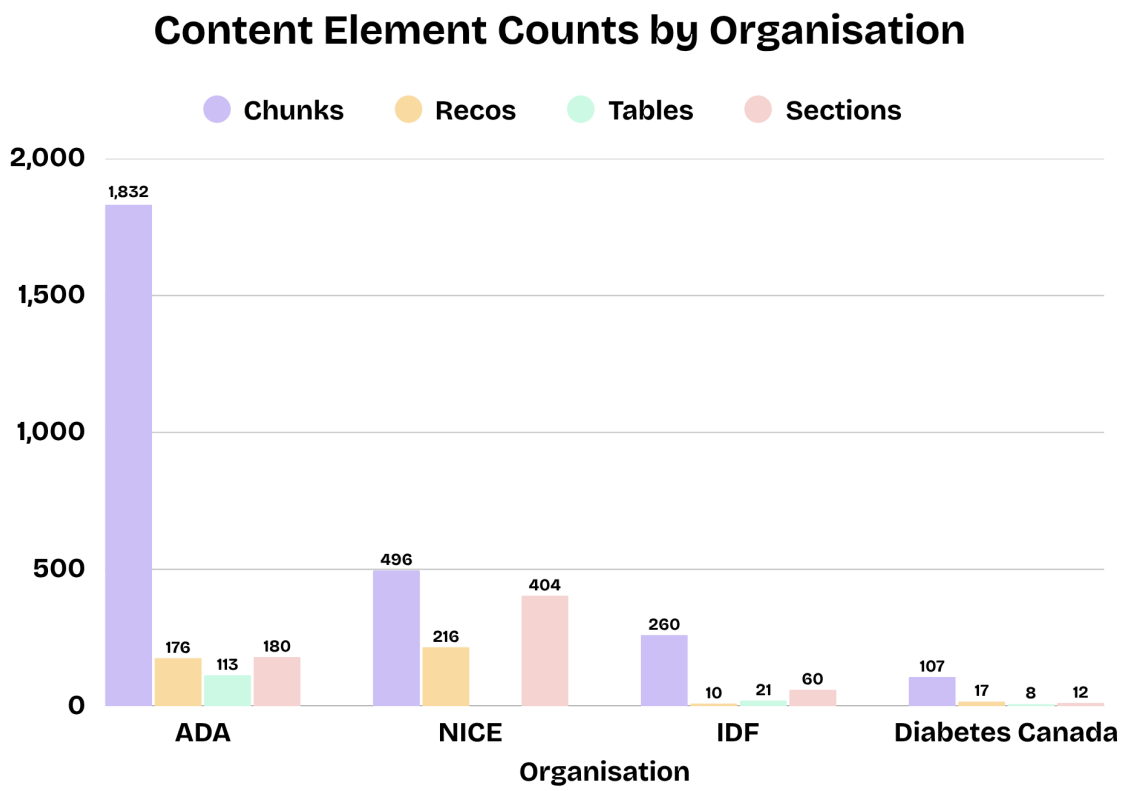


Figure 4.1.: Structural profile of the guideline corpus by organisation.

4.2. Document Ingestion and Preprocessing

The ingestion stage applies $\varphi : \mathcal{D} \to 2^{\mathcal{E}}$ to each of the eight corpus documents. This section describes the parsing tool, the document-family-specific text repairs required before extraction could begin, and the resulting element counts.

4.2.1. Document Parsing Tool

The ingestion pipeline uses PyMuPDF Lieder (2026)[5] (via the `fitz` interface) as its primary PDF extraction engine and `pdfplumber` Yang et al. (2025)[6] for table extraction.

[5]PyMuPDF, version 1.28.0, https://pymupdf.readthedocs.io.
[6]pdfplumber, version 0.11.10, https://github.com/jsvine/pdfplumber.

Docling Auer et al. (2024)[7] is retained in the ingestion script as an optional conversion step, but it was not adopted. Its output is not used by the downstream stages, and the element records are unchanged when it is unavailable. On five pages of one Diabetes Canada chapter, block extraction with PyMuPDF took 0.03 seconds and table extraction with pdfplumber 0.89 seconds (mean of five runs), against 114 seconds for a Docling conversion of the same pages on the development laptop (CPU only),[8] The text export requested by the script also carries no page numbers or coordinates. Every element record needs a page range, and reading order and heading recovery need block coordinates, so the element records are built from PyMuPDF blocks and pdfplumber tables. No systematic comparison of the parsers was carried out.

PDF extraction from scientific documents introduces known artefacts, including ligature encoding, token-boundary errors, and multi-column layout bleed Bast and Korzen (2017). PyMuPDF is the base of the pipeline because several corpus-specific steps need direct access to raw text blocks with page identifiers, font sizes, and bounding boxes. Reading order and heading recovery use the block coordinates and font sizes, and the ADA and IDF recommendation extractors work on the ordered raw blocks rather than on merged section text. A repair for concatenated words, written for the Diabetes Canada PDFs, is applied to every extracted block and to the table cells of journal-style documents. Tables are reconstructed with `pdfplumber`.

The output for each document is a set of typed JSONL files: `chunks.jsonl` for PROSE elements, `recommendations.jsonl` for RECOMMENDATION elements, a `tables/` directory of CSV files for TABLE elements, and `sections.jsonl` for SECTION elements. These constitute the intermediate representation consumed by the extraction stage.

4.2.2. Text Repair Passes

The ingestion implementation contains several document-family-specific repairs and extraction rules that were added after inspecting failures in the raw PDF output. Table 4.2 summarises the ones that directly affected the corpus representation used here.

Fix A is the most structurally significant for the Canada documents. The *Canadian Journal of Diabetes* typesetting pipeline produces PDF text streams in which word boundaries are not encoded as whitespace: the string `Ageofonset` must be reconstructed as `Age of onset` before any pattern matching is possible. The repair applies unconditional digit–letter splitting followed by conditional camelCase splitting when the density of lowercase–uppercase transitions in a block exceeds 4%.

Fix C and Fix D reflect the two most common extraction failures during development: ADA clinical directives were being silently swallowed into prose chunks, and NICE table captions were being misclassified as section headings, producing spurious SECTION elements at the start of tabular content.

No character- or token-level repairs were required for the IDF body text, whose Unicode streams are clean. Fix E instead addresses structural recognition of care-tier recommendation blocks.

[7]Docling, version 2.130.0, https://github.com/docling-project/docling.
[8]CPU 1.70 GHz, 12 cores; RAM 16GB; Intel UHD Graphics.

Fix	Artefact	Affected families	Repair operation
A	Word concatenation and camelCase runs (e.g. `Ageofonset`, `Type1diabetes`)	Diabetes Canada	Split at digit–letter and camelCase boundaries when concatenation density exceeds a threshold
B	Canada numbered recommendations parsed as prose (`"1. Diabetes should..."`)	Diabetes Canada	Dedicated regex `CANADA_RECO_RE` promotes matched blocks to RECOMMENDATION elements
C	ADA boxed recommendation text fused into single spans by PyMuPDF	ADA	Separate extractor `extract_ada_recommendation_blocks` runs on raw blocks before section merge
D	NICE table captions consumed as section headings	NICE	`looks_like_heading()` short-circuits on `NICE_TABLEISH_RE` matches when NICE numbering is active
E	IDF "Optimal / Basic / Recommended Care" prose blocks not matched as recommendations	IDF	`extract_idf_recommendations()` with pattern `IDF_CARE_RE` promotes matching blocks
F	Duplicate section headings or repeated heading blocks in the extracted page content	ADA, NICE, IDF	Per-page deduplication by title string in `recover_headings()`

Table 4.2.: Document-family-specific text repair passes.

4.2.3. Ingestion Statistics

Table 4.3 reports element counts after all repairs: the corpus produces 2,695 chunks, 419 recommendation elements, and 142 tables, dominated by the ADA document at 1,832 chunks and 176 recommendations, while the four Diabetes Canada chapters, each covering only a few pages, contribute the fewest elements individually. The 3,256 extracted content records represented by chunks, recommendations, and tables form the principal inputs used by the downstream extraction stages; the 656 section elements are consumed internally for provenance annotation and are not passed forward.

The ADA produces the largest chunk count (1,832) because its clinical directives are embedded in 377 pages of dense prose alongside structured recommendation elements. The

Org.	Document	Chunks	Recos	Tables	Sections
ADA	Standards of Care 2026	1,832	176	113	180
NICE	NG17: Type 1 Diabetes	190	100	0	167
NICE	NG28: Type 2 Diabetes	306	116	0	237
IDF	Global CPR 2025	260	10	21	60
Diab. Canada	CPG 2018 Ch. 3	30	2	5	3
Diab. Canada	CPG 2018 Ch. 4	19	2	1	3
Diab. Canada	CPG 2018 Ch. 8	23	5	1	3
Diab. Canada	CPG 2018 Ch. 9	35	8	1	3
Total		2,695	419	142	656

Table 4.3.: Content-element counts after preprocessing.

Canada chapters produce relatively small chunk and recommendation sets because each chapter is short, but both prose chunks and explicitly extracted recommendation elements remain present in the intermediate representation. The IDF produces 10 recommendation elements through its dedicated care-level recommendation extractor, alongside 21 reconstructed tables.

4.3. Knowledge Graph Construction

This section describes how the 3,256 content elements were converted into the final Knowledge Graph G, covering entity and relation extraction, graph database selection, entity deduplication, unit normalisation, cross-guideline equivalence edges, and the resulting graph statistics.

4.3.1. Entity and Relation Extraction

Extraction is implemented as two scripts that follow the split defined in Chapter 3. Extraction-derived entities carry an `extract_method` field where applicable.

Rule-based extraction. The rule extractor runs six passes in a fixed order. It first creates the Guideline and Organisation nodes and the five canonical Condition nodes. It then extracts Recommendation nodes from the numbered recommendation records produced during ingestion, numerical targets from chunk text, and table-derived nodes from the reconstructed tables. The last pass creates the equivalent_to relations once all documents have been processed. The final graph contains 70 nodes labelled `rule`: 46 GlycaemicTarget, 14 MonitoringProtocol and 10 TreatmentRecommendation. A further 491 nodes are created by the same script without a method label: 330 Recommendation, 139 Table, 8 Guideline, 4 Organisation, and 5 each of Condition and Test. A numerical target is linked to Type 1 or Type 2 diabetes only when the surrounding text names exactly one of them.

Contextual extraction. The contextual extractor sends text chunks to a language model together with the schema prompt shown in Appendix A.1. The reported graph was produced with [[gpt-4o-mini through the OpenAI API]][9], using temperature 0 and a JSON response format. For each page, the prompt also contains up to 20 facts already found by the rule extractor (numerical targets and short recommendation summaries) as known context. The parser reads the JSON response but does not check returned types against the schema. In the raw output, all 2,275 entities use declared types, but 2 of the 1,533 relations use undeclared types (RESEARCHING and INTERVENTION, one each); neither is part of the final graph. Deduplication reduces the 2,275 entities to 1,942: DRUG 627 to 438, CLINICALCONDITION 574 to 462 and INTERVENTION 337 to 305, while POPULATION stays at 737. All 1,942 carry the same confidence value of 0.7, so this field does not separate easy from hard extractions.

4.3.2. Graph Database

The Knowledge Graph is stored in Neo4j Barrasa and Webber (2023)[10] . Neo4j was chosen because it stores both nodes and edges as first-class objects with arbitrary key-value properties, which directly instantiates the property function $\lambda : V \cup E \to 2^{\mathcal{P}}$. Provenance information such as the issuing organisation and source page is stored directly on relations where it is available, without requiring separate provenance nodes. Neo4j's Cypher query language also allows the cross-guideline queries that motivate the graph to be expressed directly as graph patterns. For example, the question *which drugs are recommended as first-line therapy by more than one organisation?* can be answered by matching FIRST_LINE_FOR relations and grouping the results by organisation.

4.3.3. Entity Deduplication

Deduplication applies the type-specific equivalence relation $\sim_\tau$ defined in Section 3.6 with two strategies.

For global node types (DRUG, CONDITION, CLINICALCONDITION, INTERVENTION, TEST), two records are equivalent when their normalised names satisfy $\sigma(\text{name}_i) = \sigma(\text{name}_j)$, where σ applies whitespace normalisation, lower-casing, and type-specific synonym resolution. The implementation uses separate alias mappings for tests, drugs, and clinical conditions. Representative entries are shown in Table 4.4, which maps surface forms such as `fpg` and `fasting plasma glucose` to the canonical term `FPG`, and similarly collapses variant spellings of HbA1c and OGTT into single canonical entities.

Deduplication of this kind is studied in the entity resolution literature Getoor and Machanavajjhala (2012); Christen (2012). Clinical concept normalisation draws on the same principle as controlled vocabulary resources such as the UMLS Bodenreider (2004) and RxNorm Hanna et al. (2013), which map surface forms to canonical identifiers; the synonym dictionary Δ plays an analogous role within the scope of the corpus.

[9] Model identifier `gpt-4o-mini`, accessed through the OpenAI API, https://platform.openai.com/docs/models.

[10] Neo4j [2026.04.0], https://neo4j.com.

Surface form	Canonical form
`fpg`, `fasting plasma glucose`, `fasting glucose`	`FPG`
`hba1c`, `a1c`, `glycated haemoglobin`	`HbA1c`
`ogtt`, `2h ogtt`, `2-h ogtt`, `75g ogtt`	`2h-OGTT`

Table 4.4.: Selected synonym mappings for entity deduplication.

Organisation-specific entities retain their source scope and use type-specific identity keys rather than one common deduplication rule. GlycaemicTarget records are keyed by metric, unit, comparison operator, and value. Recommendation records use their recommendation label and text, while Table and MonitoringProtocol records use identifiers derived from the corresponding table or monitoring source. Records from different organisations are kept separate. For GlycaemicTarget records specifically, matching normalised thresholds across organisations are represented by an equivalent_to edge as described below.

4.3.4. Unit Normalisation and Cross-Guideline Equivalence Edges

All plasma glucose threshold values are normalised to mmol/L using:

$$\text{value}_{\text{mmol/L}} = \frac{\text{value}_{\text{mg/dL}}}{18.0}$$

The divisor 18.0 is the standard factor for converting plasma glucose between mg/dL and mmol/L. The pipeline's corpus values confirm this: the ADA states 126 mg/dL alongside $\geq$ 7.0 mmol/L, and 200 mg/dL alongside $\geq$ 11.1 mmol/L, American Diabetes Association Professional Practice Committee for Diabetes (2026) both consistent with the divisor. The conversion is implemented in the shared helper function `mgdl_to_mmolL`, imported by the rule extraction script. HbA1c values are retained in their source unit, and the equivalent value on the alternate scale is stored alongside it using the IFCC reference relation Hoelzel et al. (2004). Percentage values are converted to mmol/mol and mmol/mol values to percentage, and the alternate representation is stored on the target node. The current cross-guideline equivalence procedure, however, is restricted to plasma-glucose targets represented in mmol/L.

equivalent_to edges are created inside the rule extraction stage, after all documents have been processed. The function `add_cross_guideline_equivalences` groups all GlycaemicTarget nodes by the tuple (metric, op, round(value_mmol_L, 1)) and adds directed equivalent_to edges between matching target nodes from different organisations within each group. The equivalence check is exact equality on the rounded normalised value; no tolerance band was applied. Although this operation is implemented in the rule-extraction module, it runs only after the corpus-wide target extraction and numerical normalisation pass; conceptually it therefore corresponds to the normalisation stage defined in Chapter 3.

Table 4.5 lists the eight edges this process produced: three organisations agree on an FPG threshold of 7.0 mmol/L, the same three agree on a 2h-OGTT threshold of 11.1 mmol/L,

and two further edges connect a Random PG and a lower FPG bound between single pairs of organisations.

Metric	Op.	Value	Org. A	Org. B
FPG	≥	7.0 mmol/L	ADA	Diabetes Canada
FPG	≥	7.0 mmol/L	ADA	IDF
FPG	≥	7.0 mmol/L	Diabetes Canada	IDF
2h-OGTT	≥	11.1 mmol/L	ADA	Diabetes Canada
2h-OGTT	≥	11.1 mmol/L	ADA	IDF
2h-OGTT	≥	11.1 mmol/L	Diabetes Canada	IDF
Random PG	≥	11.1 mmol/L	ADA	Diabetes Canada
FPG	<	5.6 mmol/L	ADA	IDF

Table 4.5.: Cross-guideline `equivalent_to` edges after unit normalisation.

All eight edges concern diagnostic glucose thresholds, seven expressed as a lower bound ($\geq$) and one as an upper bound for the normal range ($<$ 5.6 mmol/L). The three FPG $\geq$ 7.0 mmol/L edges form a triangle across ADA, IDF, and Diabetes Canada; under the implemented equivalence criterion their corresponding target nodes fall into the same normalised group. No NICE plasma-glucose target nodes in the corpus satisfy the implemented equivalence grouping.

Figure 4.2 shows the eight edges as a matrix grouped by threshold, making visible the two three-way agreements (FPG and 2h-OGTT) against the two single-pair edges (Random PG and the normal-range FPG bound), and the absence of NICE from every row.

4.3.5. Final Graph Statistics

Tables 4.6 and 4.7 report the composition of the final Knowledge Graph after normalisation: 2,503 nodes across 13 types and 3,485 relations across 18 types, with ClinicalCondition and Drug the largest node types and mentioned_in the largest relation type at 1,409 edges. Figure 4.3 shows how these node counts distribute across the three scoping levels: global, organisation-scoped, and structural.

The high mentioned_in count (1,409, 40.4%) reflects the provenance mechanism used to connect global entities to the guidelines in which they occur. These edges preserve cross-guideline occurrence information independently of more specific clinical relations. The next most frequent, recommended_for (551), reflects the breadth of pharmacological recommendations across the four families. The 438 Drug nodes and 462 ClinicalCondition nodes are high relative to the 5 Condition nodes. This reflects the schema design, in which a small fixed set of canonical Condition nodes is pre-seeded, while drug and clinical-condition mentions encountered during extraction may produce additional nodes before synonym resolution.

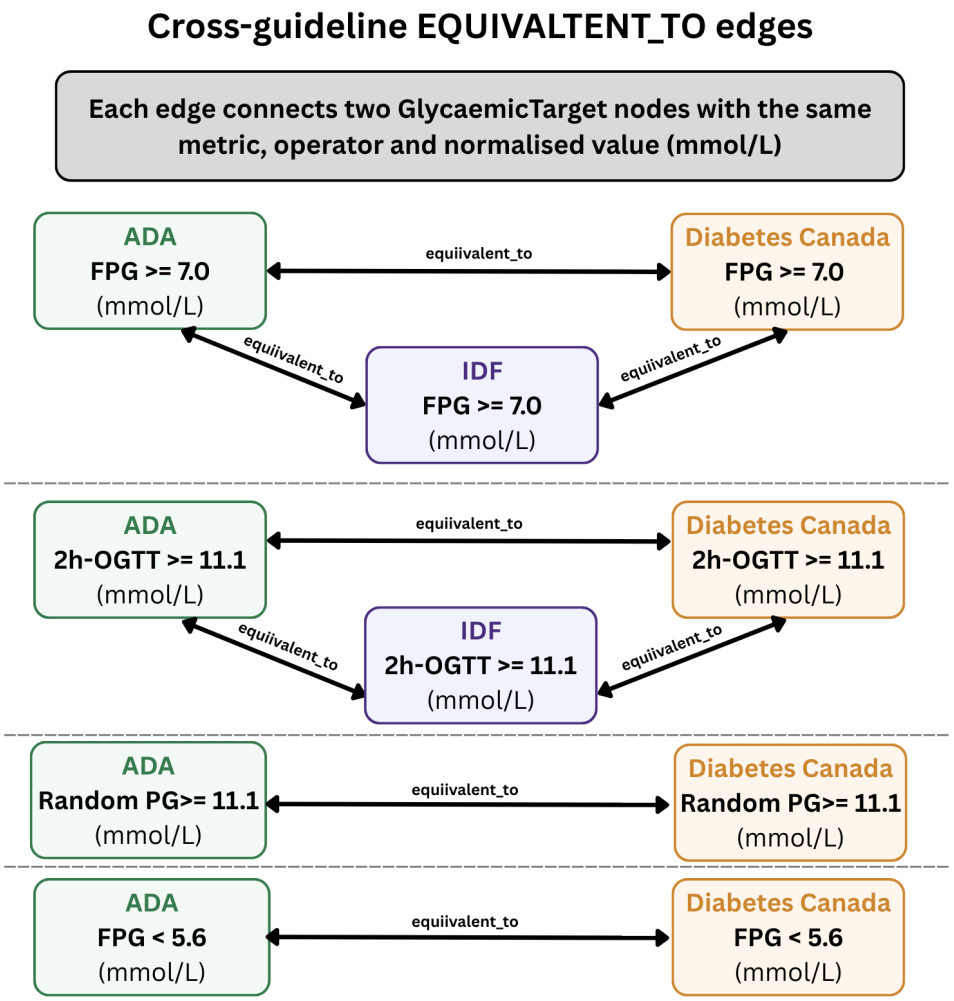


Figure 4.2.: Cross-guideline equivalence edges by diagnostic threshold.

Scoping level	Node type	Count
Global	Drug	438
	ClinicalCondition	462
	Intervention	305
	Condition	5
	Test	5
Organisation-scoped	Recommendation	330
	Table	139
	GlycaemicTarget	46
	MonitoringProtocol	14
	TreatmentRecommendation	10
Structural	Population	737
	Guideline	8
	Organisation	4
Total		**2,503**

Table 4.6.: Final knowledge graph node counts.

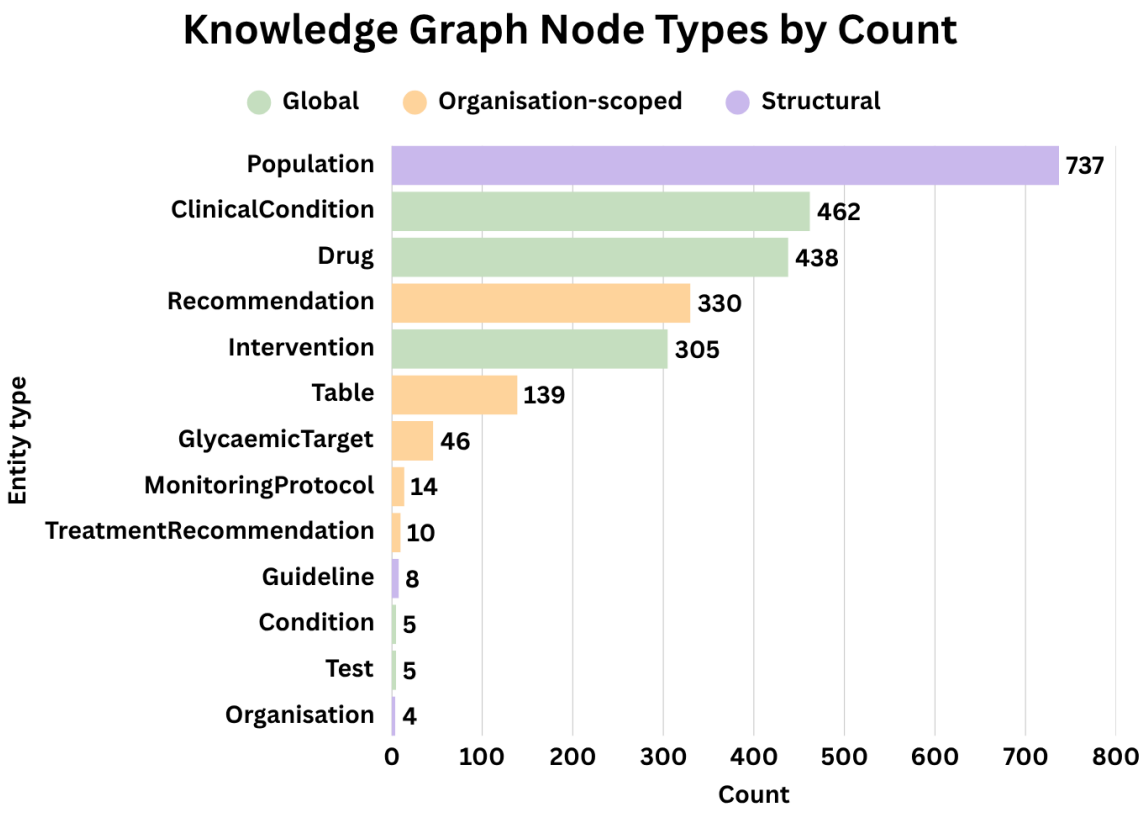


Figure 4.3.: Node-type distribution by scoping level.

To illustrate the resulting graph structure at different levels of detail, Figure 4.5 shows an example of a cross-guideline equivalence relation. Figure 4.6 focuses on the neighbourhood of the drug metformin and illustrates how a single clinical entity can be connected to multiple relations and other entities. Finally, Figure 4.4 provides an overview of the complete knowledge graph.

Condition scoping of numerical targets. The extraction of a numerical target and its assignment to a diabetes type are treated separately. A recognised numerical expression is maintained as a GlycaemicTarget regardless of whether the surrounding text identifies Type 1 or Type 2 diabetes. A disease-specific has_target relation is added only when the local context identifies one of these types unambiguously.

This distinction was introduced after an implementation check showed that the initial rule extractor used Type 2 Diabetes as the default condition when no diabetes type could be determined from the local context. This produced one has_target relation for each of the 46 extracted GlycaemicTarget nodes, even when the source text did not support a Type 2-specific assignment. The condition-assignment rule was therefore changed so that ambiguous targets remain in the graph without a disease-specific relation.

With this correction, all 46 GlycaemicTarget nodes are retained, while 24 has_target relations are created. Seven originate from the Type 1 Diabetes node and 17 from the

Semantic group	Relation type	Count
Pharmacological	RECOMMENDED_FOR	551
	MANAGES	272
	FIRST_LINE_FOR	173
	ASSOCIATED_WITH	157
	CONTRAINDICATED_IN	110
	SECOND_LINE_FOR	69
	USED_WITH	67
Monitoring & measurement	REQUIRES_MONITORING	53
	MEASURED_BY	46
	HAS_TARGET	24
	SPECIFIES_TARGET	32
	DESCRIBES_PROTOCOL	23
	HAS_PROTOCOL	4
Cross-guideline & provenance	MENTIONED_IN	1,409
	HAS_RECOMMENDATION	340
	CONTAINS_TABLE	139
	PUBLISHES	8
	EQUIVALENT_TO	8
Total		**3,485**

Table 4.7.: Final knowledge graph relation counts.

Type 2 Diabetes node. Some targets can be associated with both conditions when separate source contexts support both assignments. The eight cross-guideline EQUIVALENT_TO relations are unaffected because their construction depends on the normalised target properties rather than on HAS_TARGET.

4.4. Question Generation

The question generation stage applies the three-function pipeline $\gamma = \nu \circ \beta \circ \alpha$ defined in Chapter 3 to the final Knowledge Graph, where α is the graph-traversal targeting function, β is the language-model formulation function, and ν is the rule-based validation function. In other words, a graph edge is first selected as an anchor by α, converted into question text by β, and then checked against the same anchor by ν. This follows the schema-constrained prompting approach of Caufield et al. (2024) and produces a validated catalogue $\mathcal{Q}^+$.

Two question-generation runs are reported in this chapter and are analysed separately. The first run used Claude Opus 4.8 [11] through the Anthropic API and was stopped before all eligible graph anchors had been processed because of practical token and cost constraints. This partial run produced a catalogue of 491 validated questions and was the source of the questions used in the human clinical review in Section 4.5.4. A subse-

[11] Model identifier `claude-opus-4-8`, accessed through the Anthropic API, https://docs.anthropic.com;

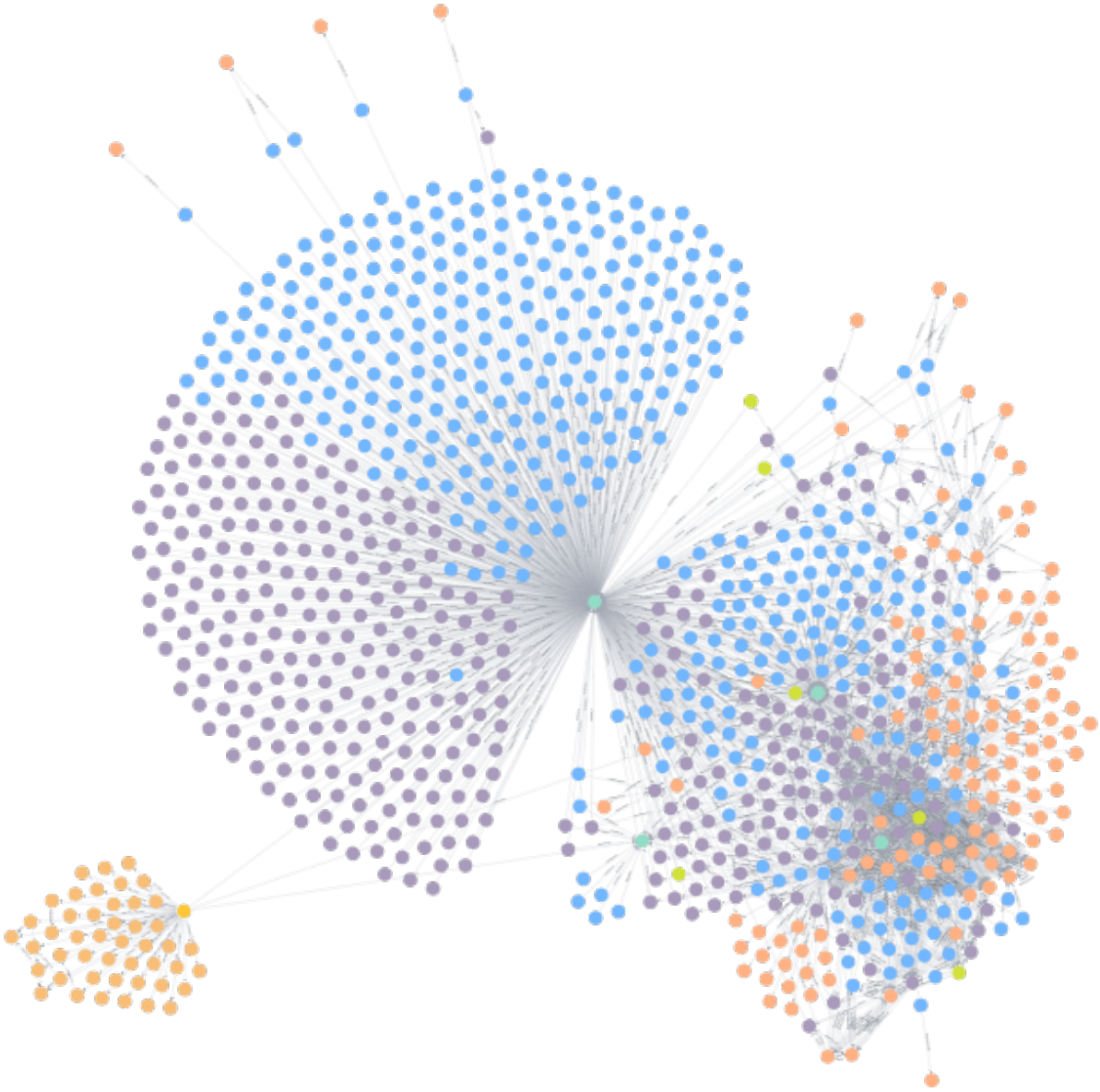


Figure 4.4.: Overview of the complete knowledge graph.

quent run used gpt-4o-mini [12], using the same final Knowledge Graph and provides the catalogue-level generation and validator statistics reported in this section. Results from the two runs are not pooled.

4.4.1. Catalogue Overview

In the GPT-4o-mini run, the targeting and generation pipeline produced 1,289 question records from eligible graph anchors. The rule-based validator accepted 1,280 questions (99.3%) into $\mathcal{Q}^+$ and routed 9 questions (0.7%) to the flagged catalogue $\mathcal{Q}^-$. No LLM

[12]Model identifier `gpt-4o-mini`, accessed through the OpenAI API, https://platform.openai.com/docs/models.

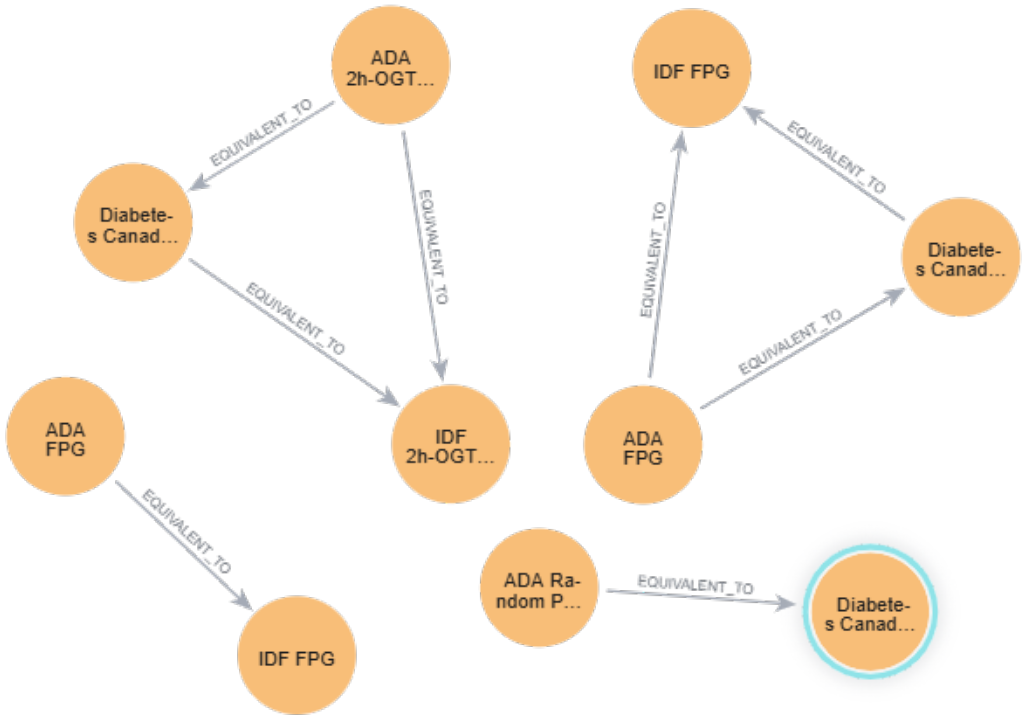


Figure 4.5.: Example of a cross-guideline equivalence relation in the knowledge graph.

fallback was required during this run. Of the 1,289 generated questions, 296 (23.0%) were successfully produced with multiple-choice options, using LLM-generated distractors and a graph-locked correct answer. Of these, 294 were retained in the accepted catalogue after validation.

4.4.2. Distribution by Relation Type

Relation-type-driven question generation is an established approach in knowledge-graph question generation. The distribution reflects both the graph edge frequency and the per-type sampling limits in α. Table 4.8 shows the question counts per relation type: RECOMMENDED_FOR produces the most questions at 383, followed by MANAGES at 265, while EQUIVALENT_TO, despite drawing on only eight graph edges, still produces 8 cross-guideline questions that no single-guideline source could support.

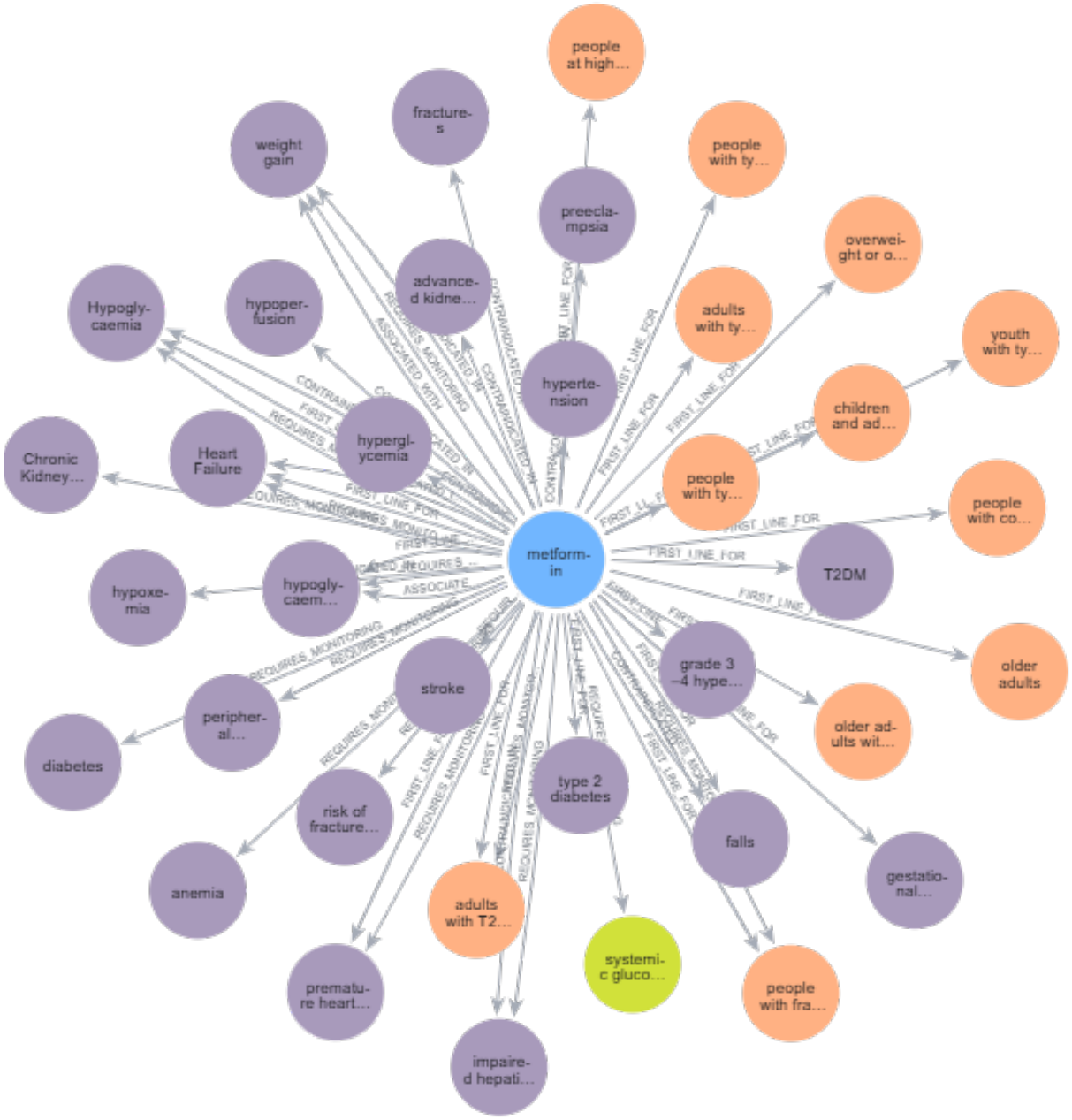


Figure 4.6.: Example of the graph neighbourhood of the drug metformin.

Figure 4.7 shows how the generated questions were distributed across the graph relation types and the three primary generation methods. The distribution reflects the fixed relation-to-method mapping used by the question-generation pipeline. Grounded prompts were used for relations where the answer could be directly formulated from the graph assertion, while clinical vignettes and template paraphrases were used for the corresponding relation types defined by the generation configuration.

Figure 4.8 shows the distribution of accepted questions across the four question types and the three primary generation methods. The figure also indicates which accepted questions

Relation type	Questions	Question form
RECOMMENDED__FOR	383	*which*
MANAGES	265	*how* (vignette)
MENTIONED__IN	125	*which* (paraphrase)
FIRST__LINE__FOR	117	*which* + MCQ
ASSOCIATED__WITH	107	*what*
CONTRAINDICATED__IN	107	*yes/no* + MCQ
SECOND__LINE__FOR	51	*which* + MCQ
USED__WITH	60	*what*
REQUIRES__MONITORING	45	*what*
HAS__TARGET	21	*what* + MCQ
EQUIVALENT__TO	8	*how*
Total	**1,289**	

Table 4.8.: Questions by relation type and question form.

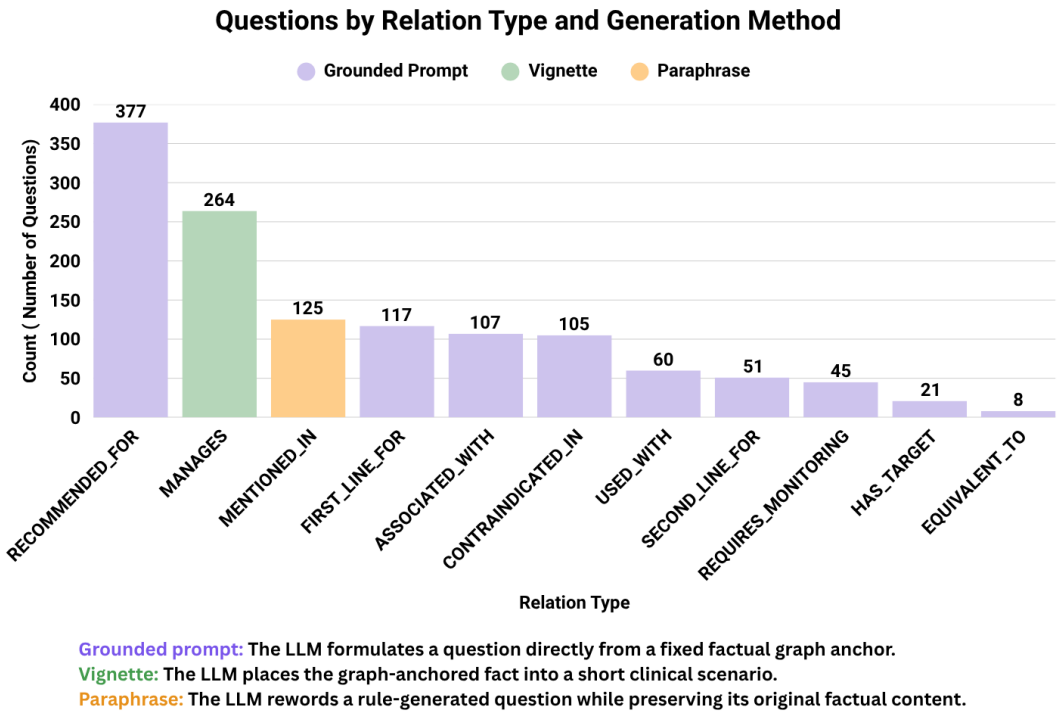


Figure 4.7.: Question distribution by relation type and generation method.

were produced in multiple-choice format (MCQ). MCQ is treated as an output format rather than a separate generation method.

4.4.3. Distribution by Generation Method and Question Type

The which form is the most frequent (52.4%), driven primarily by RECOMMENDED__FOR, FIRST__LINE__FOR, SECOND__LINE__FOR, and the cross-guideline MENTIONED__IN ques-

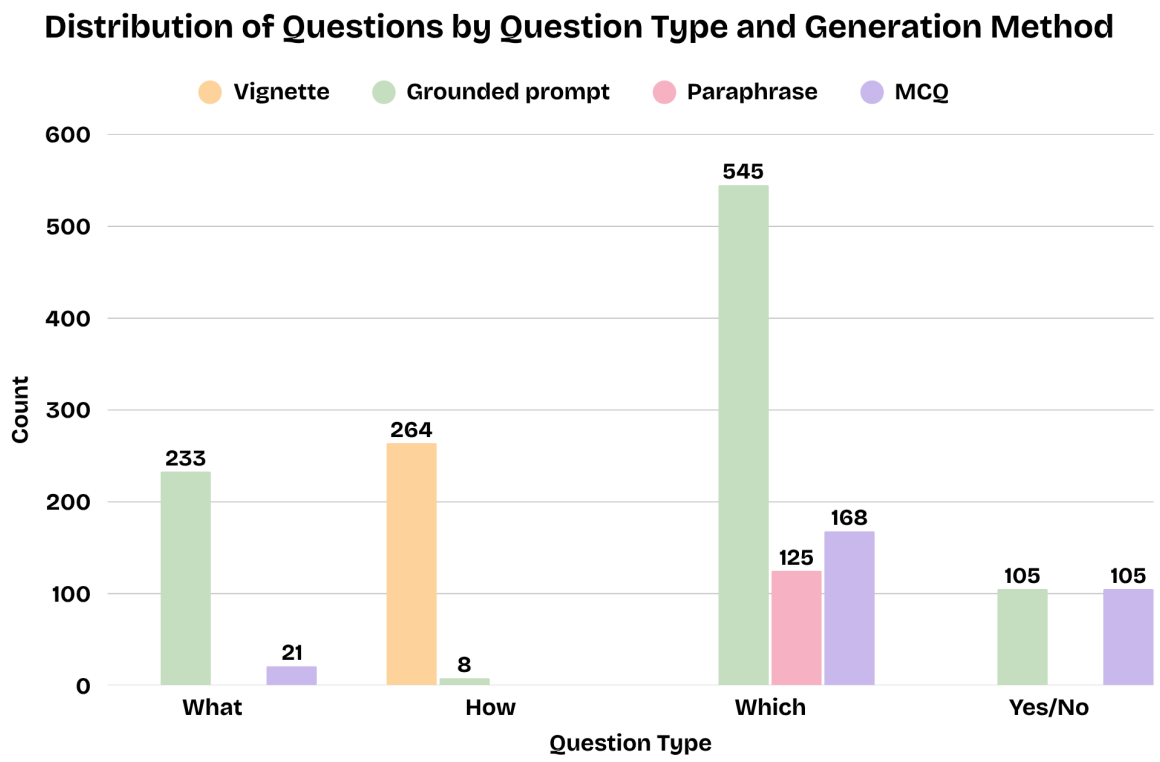


Figure 4.8.: Distribution of accepted questions by question type, primary generation method, and MCQ format.

Method	N	%	Type	N	%
Graph-grounded prompt	899	69.7	*which*	676	52.4
Clinical vignette	265	20.6	*how*	273	21.2
Template paraphrase	125	9.7	*what*	233	18.1
			yes/no	107	8.3
Total	1,289	100	**Total**	1,289	100

Table 4.9.: Questions by generation method and question type in the full GPT-4o-mini catalogue.

tions. The how form accounts for 21.2% and is dominated by the vignette-based MANAGES relations. The yes/no form remains the least frequent (8.3%) because it is restricted to CONTRAINDICATED_IN edges.

4.4.4. Cross-Guideline Questions

The eight questions derived from EQUIVALENT_TO anchors are the only ones in the catalogue requiring knowledge spanning two guidelines. Each anchor pairs two GLYCAEMIC-

TARGET nodes from different organisations that share the same normalised threshold. An example question from the FPG $\geq$ 7.0 mmol/L edge between ADA and Diabetes Canada:

> *What fasting plasma glucose threshold is shared by the ADA and Diabetes Canada target nodes, and in what units does each guideline state it?*

This question type is not possible from a single-guideline pipeline in this corpus or by extraction from raw text, because it depends on an EQUIVALENT_TO edge that does not exist in any source document.

4.4.5. Representative Examples

Table 4.10 shows representative questions drawn from the validated catalogue, grouped by generation method and question type. The examples illustrate the linguistic variety the hybrid pipeline achieves compared to a single fixed template: the grounded-prompt method produces direct factual questions; the vignette method produces clinical scenario questions that embed the same pharmacological fact in a patient narrative (the patient details, such as age group, sex or symptoms, are added by the model and are not drawn from the anchor); and questions derived from EQUIVALENT_TO anchors require the student to compare positions across two guidelines simultaneously.

Method	**Type**	**Question**
Graph-grounded prompt	*which* (recall)	Which pharmacological approach is recommended for managing blood glucose in people with diabetes who are at higher risk of hypoglycaemia?
Graph-grounded prompt	*what* (recall)	What fasting plasma glucose level is used as the threshold for identifying individuals at risk of developing type 2 diabetes?
Graph-grounded prompt + MCQ	*what* (recall)	In a patient presenting with classic symptoms of hyperglycaemia, what random plasma glucose value meets the diagnostic threshold for diabetes mellitus? *A.* $\geq$ 180.0 mg/dL *B.* $\geq$ 140.0 mg/dL *C.* $\geq$ 200.0 mg/dL *D.* $\geq$ 160.0 mg/dL
Clinical vignette	*how* (application)	A middle-aged man with a history of type 2 diabetes and recently diagnosed stable heart failure presents for a routine medication review. He reports mild ankle swelling and reduced exercise tolerance. Which pharmacological management strategy should be prioritised given his dual diagnosis?
Cross-guideline	*how* (application)	How does the fasting plasma glucose threshold used to diagnose diabetes compare between the American Diabetes Association and Diabetes Canada guidelines?

Table 4.10.: Representative questions from the validated catalogue.

4.5. Evaluation of Question Quality

This section evaluates question quality beyond the automatic validator. A rule-based baseline is used to compare output variety, lexical diversity is assessed at catalogue level, and a sample from the partial Claude Opus 4.8 run is examined through independent clinical review.

4.5.1. Baseline Definition

The baseline for comparison is a rule-only question generator that produces one canonical question per anchor triple using a fixed directional template per relation type. For a RECOMMENDED_FOR edge with source v_s and destination v_d, the canonical template produces: *"Which drug is recommended for v_d?"* with the answer fixed to v_s. No language model is called, no reformulations are generated, and no MCQ distractors are produced. Anchor consistency holds trivially here, because the question text contains only the node names from the anchor. Hallucination in language models is a well-documented problem Huang et al. (2025); the validator is designed specifically to catch faithfulness failures where the generated text departs from the supplied anchor fact. The baseline represents the minimum viable question catalogue: graph-anchored, consistently structured, and producible without any LLM calls or additional cost.

The hybrid pipeline, described in Section 4.4, differs from this baseline in three respects: the surface form of each question is generated from an anchor-restricted prompt; reformulations of the question are generated at up to three distinct phrasings; and multiple-choice distractors are generated for selected relation types. The rule-only baseline is anchor-consistent by construction because its question and answer are assembled directly from the graph anchor. For the hybrid pipeline, anchor consistency is checked after language-model formulation by the rule-based validator, and only questions passing these checks enter the validated catalogue. Neither case implies that the underlying graph edge is clinically correct.

4.5.2. Validator Analysis

The validator applies three checks to every generated question. Check 1 verifies that the correct answer appears in the question text or is recoverable as a named graph entity. Check 2 flags any number in the generated question that was not present in the anchor's node properties (with small integers 0–4 whitelisted as non-clinical). Check 3 verifies that if a specific guideline is named in the question, it matches the anchor's issuing organisation.

Nine questions were routed to $\mathcal{Q}^-$, all because Check 2 detected numerical information in the generated question that was not represented among the numerical properties permitted by the validator. Six originated from RECOMMENDED_FOR relations, two from CONTRAINDICATED_IN, and one from a MANAGES relation. The flagged values included patient ages or age ranges, a postpartum duration, an eGFR range, and an age introduced in a vignette. Inspection showed that several of these values reflected information embedded in population or condition descriptions rather than newly invented clinical thresholds.

The numerical validator is therefore deliberately conservative and can flag contextually appropriate numbers when they are not represented in the specific anchor properties it checks. The resulting 99.3% acceptance rate measures consistency with the implemented validation rules rather than clinical accuracy.

4.5.3. Automatic Metrics

Table 4.11 compares the hybrid pipeline against the rule-only baseline: the hybrid pipeline generates 294 multiple-choice questions and 3.38 phrasings per anchor on average, against none and exactly 1.00 for the rule-only baseline. The validator pass rate is lower for the hybrid pipeline (99.3% against 100%) for the reasons discussed in Section 4.5.2.

Metric	Rule-only baseline	Hybrid pipeline
Validator pass rate	100%	99.3%
Questions with MCQ	0	294
Mean phrasings/anchor	1.00	3.38
MSTTR (window 50)	n/a	0.716
Primary difficulty labels	n/a	2 (recall, application)

Table 4.11.: Automatic metrics for the rule-only and hybrid pipelines.

The main quantitative difference between the two approaches is the variety of question forms produced. The accepted GPT-4o-mini catalogue contains 1,280 primary questions and 3,048 stored reformulations, giving 4,328 phrasings in total. To describe this variation, the mean number of stored phrasings per accepted anchor is calculated as

$$\text{Mean phrasings per anchor} = \frac{N_{\text{primary}} + N_{\text{reformulations}}}{N_{\text{primary}}}.$$

Since each primary question corresponds to one accepted graph anchor, this gives

$$\frac{1280 + 3048}{1280} = 3.38$$

stored phrasings per anchor.

Lexical diversity is measured using the Mean Segmental Type–Token Ratio (MSTTR) Ahn et al. (2025). Unlike the conventional type–token ratio, MSTTR reduces the effect of text length by calculating lexical diversity over equal-length, non-overlapping segments:

$$\text{MSTTR} = \frac{1}{k} \sum_{i=1}^{k} \frac{t_i}{w_i},$$

where k is the number of segments, t_i is the number of unique word types in segment i, and w_i is the number of tokens in that segment. Using a 50-token window across the primary questions and their reformulations gives an MSTTR of 0.716.

The accepted catalogue also contains 294 questions with MCQ options. Primary questions are labelled as either recall or application depending on the generation method, while

the reformulations additionally include analysis variants. In comparison, the rule-only baseline produces a single phrasing per anchor, does not generate MCQ options, and does not assign difficulty labels.

4.5.4. Human Clinical Review

A sample of 30 questions from the 491-question partial Claude Opus 4.8 catalogue was reviewed for clinical soundness by two medical doctors. The questions were presented without the graph anchors that produced them, so that each reviewer judged the question on its own clinical merits rather than against the system's intended answer. The two review sets of 15 questions each were non-overlapping and sampled across generation methods in proportion to their frequency in the catalogue.

Evaluation of clinical NLP systems often relies on comparison against expert-annotated reference standards Uzuner et al. (2011), and for generated clinical text, evaluation by physicians has been used to assess quality along clinical dimensions Singhal et al. (2025). Each reviewer was asked to mark each question as clinically correct or incorrect, and to provide one line of explanation for any question marked incorrect. Both reviewers are qualified medical doctors from India and were contacted through the author's personal network. Dr. Mounika is currently in the final year of postgraduate training in Obstetrics and Gynaecology. Dr. Aakanksha completed her undergraduate medical education in India, and is currently preparing for the Fachsprachprüfung (FSP) in Heidelberg, Germany as part of the process of continuing medical practice in Germany. Both participated voluntarily and without compensation, and agreed to take part and to be described in this thesis. The review involved no patients and no personal data beyond the reviewers' professional background reported here. Each reviewed a separate set of 15 questions, so the two sets do not overlap and no question was seen by both reviewers. Reviewers worked independently and were not in contact with each other during the review. They were shown the question text only, without the graph anchor, the answer key, or the source guideline passage, so each judgement was made on the question as a learner would encounter it. The reviewers were also not informed which language model had generated the questions. Because the guidelines in the corpus are American, British, Canadian, and international, reviewers were asked to indicate where a flag reflected a difference from routine practice in their own clinical context rather than a factual error in the question.

The human review showed that most questions were considered acceptable, while several issues were identified in the reviewed sample. Table 4.12 summarises the individual review results, while Figure 4.9 shows the distribution of the review outcomes for the two reviewers. The following discussion examines the identified issues in more detail.

Of the 30 reviewed questions, 26 were accepted without qualification. Three were classified as clear failures and were traced to incorrect relations in the Knowledge Graph. One additional question (Q6) was flagged because its content, although supported by the source guidelines, was outside routine practice in the reviewer's clinical context. This case is therefore treated as a clinical-scope flag rather than a factual error.

Within this sample, all three clear failures originated from incorrect graph relations rather than from the question-formulation step. The sample is too small to establish a general

Q	Reviewer	Reviewer comment	Root cause
Q9	Dr. Aakanksha	Metformin identified as managing hypoglycaemic events; sulfonylureas are the drug class associated with hypoglycaemia	Extraction error: the source passage does not support the extracted `metformin -MANAGES-> hypoglycaemic events` relation
Q15	Dr. Aakanksha	NHS Path to Remission recommended for Heart Failure; SGLT2 inhibitors are the standard pharmacological approach for T2DM with Heart Failure	Extraction error: the source passage does not support the extracted `manages` relation between this programme and Heart Failure
Q3	Dr. Mounika	No specific antidiabetic is recommended for fractures; perioperative glycaemic control is managed by insulin dose adjustment only	Extraction error: the source passage does not support the extracted `insulin -RECOMMENDED_FOR-> fractures` relation
Q6	Dr. Mounika	Bariatric surgery for T2DM is evidence-based but not routine practice; noted as practised differently in India	Clinical scope: ADA and IDF guidelines do recommend metabolic surgery for eligible adults with T2DM; not an extraction error

Table 4.12.: Human clinical review results.

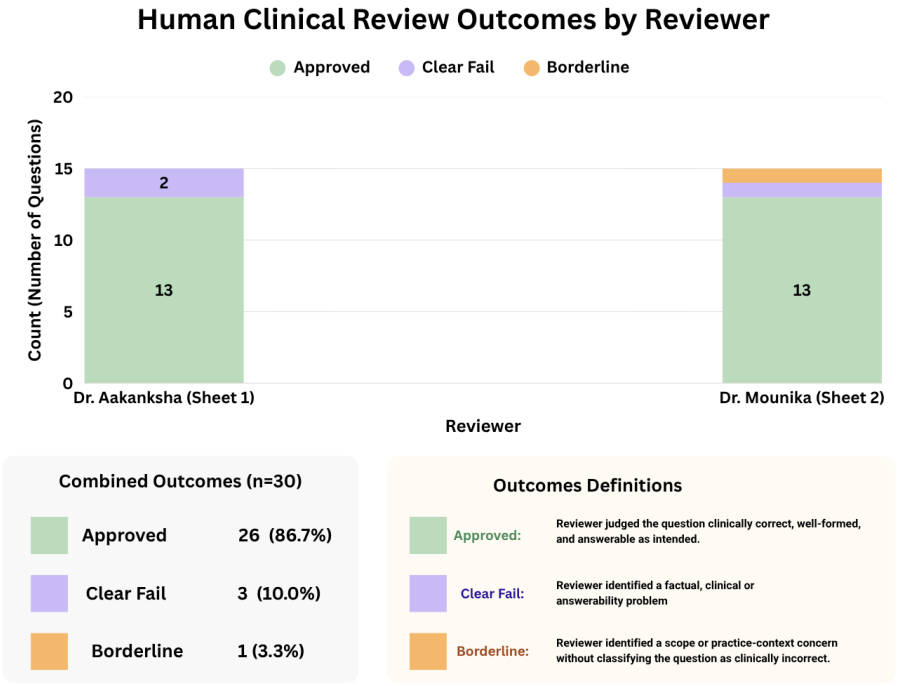


Figure 4.9.: Human clinical review outcomes by reviewer.

reliability rate, but it indicates that graph quality was the main source of factual error among the reviewed questions.

One additional question was flagged by Dr. Aakanksha (Q5), which asked how the ADA and IDF diagnostic thresholds for the 2-hour oral glucose tolerance test compare. The reviewer noted that both guidelines use the same threshold of $\geq$ 11.1 mmol/L, which is the intended answer to the question. This confirms that the cross-guideline comparison question type produced from EQUIVALENT_TO edges is answerable. It is not counted as a failure.

4.5.5. Comparison with Baseline

Clinical correctness against the original guideline sources was not independently measured for the rule-only baseline, so the comparison is limited to properties of the generated catalogue rather than clinical accuracy.The accepted hybrid catalogue contains 1,280 primary questions together with 3,048 additional reformulations and 294 questions with MCQ options. Primary questions carry recall or application difficulty labels, while the stored reformulations additionally include recall, application, and analysis variants. This provides a wider range of question forms than the fixed single-phrasing rule-only baseline. Whether this additional variety translates into improved learning outcomes is not evaluated in this thesis and remains a question for future study.

4.6. Limitations

The experiments and the final implementation showed several limitations that would need to be addressed in further work.

Entity matching. Drug and clinical-condition names are merged using fixed alias lists. This works for the variants included in those lists, but an alternative name that is not listed can remain as a separate node. The current implementation therefore depends on knowing common naming variations in advance. Extending the alias lists would improve coverage, although the additional mappings would need to be checked to avoid merging different clinical concepts.

Cross-guideline target comparison. The current EQUIVALENT_TO comparison is limited to plasma-glucose targets that have a value in mmol/L. HbA1c targets are also extracted, and conversions between percentage and mmol/mol are stored, but these values are not included in the equivalence comparison. The eight equivalence relations reported in this chapter therefore cover only the plasma-glucose targets matched by the implemented criterion.

Condition scoping. The diabetes type of a numerical target is determined from the text in which the target is extracted. If that text mentions both Type 1 and Type 2 diabetes, or does not identify either one, the target is kept but no disease-specific HAS_TARGET relation is added. This is safer than assigning a default condition, but some targets

consequently remain without a Type 1 or Type 2 link. Section or document context could be used in addition to the local text in a future version.

Contextual extraction. Contextual entity and relation extraction contributes directly to the final Knowledge Graph used by both question-generation runs. Although the LLM extractor is constrained through the prompt to predefined entity and relation types and is instructed not to introduce relations unsupported by the source passage, these constraints are not enforced independently against the original guideline after extraction. Errors introduced during contextual extraction can propagate into the final graph and subsequently into generated questions. This was observed during human review, where some clinically questionable questions remained consistent with incorrect or questionable graph relations. Since the question validator checks graph-anchor consistency rather than independently verifying the source guideline, such upstream errors are outside its scope. A systematic source-level evaluation of contextual extraction was not carried out and remains future work.

Generalisability. The pipeline was evaluated only on the diabetes guideline corpus used in this thesis. Several ingestion rules are specific to the four publishing formats in the corpus. For example, the implementation contains separate handling for ADA recommendation blocks, NICE document structure, IDF care-level recommendations, and Diabetes Canada text extraction. A new guideline family may therefore require additional document-specific handling before the rest of the pipeline can be applied.

5. Conclusions and Future Work

5.1. Conclusion and Key Findings

This thesis addressed a practical problem: clinical practice guidelines from different organisations are distributed across heterogeneous documents and are not directly available in a common representation for cross-guideline querying and comparison. The approach taken was to construct a knowledge graph that represents content from multiple guidelines in a shared structure and then use that graph as a controlled source for question generation.

The four-stage pipeline: ingestion, extraction, normalisation, and question generation, was designed to handle the structural and terminological variation that arises when documents from different publishing organisations are processed together. The ingestion stage classified document content into typed element records while preserving page-level provenance. The extraction architecture separates entity types between a rule-based layer and a contextual layer, with the full design supporting LLM-based extraction for categories that require contextual interpretation. The normalisation stage resolved duplicate entities, converted measurement units to a common representation, and created equivalence edges between matching targets from different organisations. The question-generation stage selected graph edges as anchors, provided each anchor to the language model as factual context, and checked the generated question for consistency with that anchor before acceptance.

The pipeline was instantiated on eight documents from four organisations covering Type 1 and Type 2 diabetes. The resulting knowledge graph contains 2,503 nodes across 13 node types and 3,485 edges across 18 relation types. Normalisation created eight EQUIVALENT_TO edges linking matching diagnostic thresholds across ADA, IDF, and Diabetes Canada after unit conversion. These relations are not directly stated in any individual source document; they arise from comparing independently stated targets after normalisation.

The full GPT-4o-mini question-generation run produced 1,289 questions, of which 1,280 passed the rule-based validation checks and 9 were flagged by the rule-based validator because they contained numerical values absent from their anchors. A separate 491-question catalogue was used for the detailed quality evaluation. From this catalogue, 30 questions were reviewed by two medical doctors: 26 were accepted without qualification, three contained clear extraction-derived errors, and one was flagged for a clinical-scope concern rather than a factual error.

In the reviewed sample, all three clear failures were traced to incorrect graph relations rather than to the question-formulation step. This points to graph quality, particularly the correctness of extracted relations, as an important target for further improvement.

5.2. Future Work

Several directions follow from the work reported here.

The ingestion pipeline evaluated Docling as an alternative to PyMuPDF but did not adopt it, primarily because of its runtime cost on the CPU-only development hardware (Section 4.2.1). A spot check on a small sample showed Docling produced no ligature artefacts, unlike the PyMuPDF output, though the comparison of word-spacing errors between the two was inconclusive. With GPU acceleration available, re-evaluating Docling as the primary parser, particularly for its ligature handling, would be a reasonable next step, and could reduce or replace the family-specific text repairs currently required for some document sources.

The LLM-based contextual extraction path was executed for the four entity categories of drugs, clinical conditions, populations, and interventions. However, extraction quality was not independently evaluated against a manually annotated gold-standard dataset, and the LLM-based extraction was not compared with the heuristic fallback mode of the same script.

Entity deduplication in this work is deterministic and relies on a fixed synonym dictionary. Surface forms not represented in the dictionary remain as separate nodes, which can increase the number of near-duplicate Drug and ClinicalCondition nodes. Embedding-based or fuzzy matching could improve recall in entity resolution, although safeguards would be needed to avoid incorrectly merging distinct entities.

The current EQUIVALENT_TO criterion is implemented for normalised plasma-glucose targets. Although alternate HbA1c representations are stored during extraction, the equivalence procedure does not currently compare HbA1c targets across organisations. Extending the criterion to HbA1c would enable additional cross-guideline comparisons, including comparisons involving NICE targets represented in the corpus.

The validator checks consistency between a generated question and its graph anchor, but it does not determine whether the underlying graph relation is clinically correct. A more complete evaluation would include a larger reviewed sample, reviewers with a broader range of relevant clinical expertise, and an analysis of whether the graph errors observed in the current review are systematic or isolated.

Finally, the four-stage architecture is not tied to diabetes, but the schema, extraction rules, canonical entities, repair procedures, and normalisation configuration used here were developed for the diabetes corpus. Applying the architecture to another multi-guideline domain would test which components transfer with limited adaptation and which require domain-specific redesign.

Bibliography

Ahn, S., Park, J., Go, G., Kim, C., Jung, J., Shin, M. S., Kim, D.-G., and Seo, Y.-D. (2025). Ukta: Unified korean text analyzer. arXiv:2502.09648. https://arxiv.org/abs/2502.09648.

American Diabetes Association Professional Practice Committee for Diabetes (2026). Introduction and methodology: Standards of care in diabetes—2026. *Diabetes Care*, 49(Supplement 1). https://doi.org/10.2337/dc26-SINT.

Anderson, L. W. and Krathwohl, D. R. (2001). *A Taxonomy for Learning, Teaching, and Assessing : A Revision of Bloom's Taxonomy of Educational Objectives : Complete Edition*. Addison Wesley Longman.

Auer, C., Lysak, M., Nassar, A., Dolfi, M., Livathinos, N., Vagenas, P., Ramis, C. B., Omenetti, M., Lindlbauer, F., Dinkla, K., Mishra, L., Kim, Y., Gupta, S., de Lima, R. T., Weber, V., Morin, L., Meijer, I., Kuropiatnyk, V., and Staar, P. W. J. (2024). Docling Technical Report. arXiv:2408.09869.

Barrasa, J. and Webber, J. (2023). *Building Knowledge Graphs*. "O'Reilly Media, Inc.".

Bast, H. and Korzen, C. (2017). A benchmark and evaluation for text extraction from pdf. In *2017 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, pages 1–10.

Berard, L. D., Siemens, R., and Woo, V. (2018). Monitoring glycemic control. *Canadian Journal of Diabetes*, 42(Supplement 1):S47–S53. https://www.diabetes.ca/for-professionals/full-guidelines/chapter-9.

Bi, S., Cheng, X., Li, Y.-F., Wang, Y., and Qi, G. (2020). Knowledge-enriched, Type-constrained and Grammar-guided Question Generation over Knowledge Bases. In Scott, D., Bel, N., and Zong, C., editors, *Proceedings of the 28th International Conference on Computational Linguistics*, pages 2776–2786, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Bodenreider, O. (2004). The Unified Medical Language System (UMLS): Integrating biomedical terminology. *Nucleic Acids Research*, 32(Database issue):D267–270.

Borchert, F., Lohr, C., Modersohn, L., Langer, T., Follmann, M., Sachs, J. P., Hahn, U., and Schapranow, M.-P. (2020). GGPONC: A corpus of German medical text with rich metadata based on clinical practice guidelines. In Holderness, E., Jimeno Yepes, A., Lavelli, A., Minard, A.-L., Pustejovsky, J., and Rinaldi, F., editors, *Proceedings of the 11th International Workshop on Health Text Mining and Information Analysis*, pages 38–48, Online. Association for Computational Linguistics. https://aclanthology.org/2020.louhi-1.5/.

Brizan, D. and Tansel, A. (2015). A. Survey of Entity Resolution and Record Linkage Methodologies. *Communications of the IIMA*, 6(3).

Caufield, J. H., Hegde, H., Emonet, V., Harris, N. L., Joachimiak, M. P., Matentzoglu, N., Kim, H., Moxon, S., Reese, J. T., Haendel, M. A., Robinson, P. N., and Mungall, C. J. (2024). Structured Prompt Interrogation and Recursive Extraction of Semantics (SPIRES): A method for populating knowledge bases using zero-shot learning. *Bioinformatics*, 40(3):btae104.

Chiticariu, L., Li, Y., and Reiss, F. R. (2013). Rule-Based Information Extraction is Dead! Long Live Rule-Based Information Extraction Systems! In Yarowsky, D., Baldwin, T., Korhonen, A., Livescu, K., and Bethard, S., editors, *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pages 827–832, Seattle, Washington, USA. Association for Computational Linguistics.

Christen, P. (2012). *Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection*. Data-Centric Systems and Applications. Springer, Berlin ; New York.

Ding, Y., Han, S. C., Lee, J., and Hovy, E. (2025). Deep Learning based Visually Rich Document Content Understanding: A Survey. arXiv:2408.01287.

Ekoe, J.-M., Goldenberg, R., and Katz, P. (2018). Screening for diabetes in adults. *Canadian Journal of Diabetes*, 42(Supplement 1):S16–S19. https://www.diabetes.ca/for-professionals/full-guidelines/chapter-4.

Färber, M., Bartscherer, F., Menne, C., and Rettinger, A. (2017). Linked data quality of DBpedia, Freebase, OpenCyc, Wikidata, and YAGO. *Semantic Web*, 9(1):77–129.

Federation, I. D. (2025). Idf global clinical practice recommendations for managing type 2 diabetes 2025. *Diabetes Research and Clinical Practice*, 224:112238. https://doi.org/10.1016/j.diabres.2025.112238.

Gao, Y. (2022). *A combined rule-based and machine learning approach for blackout analysis using natural language processing*. laurea, Politecnico di Torino.

Gao, Y., Bing, L., Li, P., King, I., and Lyu, M. R. (2019). Generating Distractors for Reading Comprehension Questions from Real Examinations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01):6423–6430.

Getoor, L. and Machanavajjhala, A. (2012). Entity resolution: Theory, practice & open challenges. *Proceedings of the VLDB Endowment*, 5(12):2018–2019.

Guo, S., Liao, L., Li, C., and Chua, T.-S. (2024). A Survey on Neural Question Generation: Methods, Applications, and Prospects. arXiv:2402.18267.

Guyatt, G., Oxman, A. D., Akl, E. A., Kunz, R., Vist, G., Brozek, J., Norris, S., Falck-Ytter, Y., Glasziou, P., deBeer, H., Jaeschke, R., Rind, D., Meerpohl, J., Dahm, P., and Schünemann, H. J. (2011). Grade guidelines: 1. introduction—grade evidence profiles and summary of findings tables. *Journal of Clinical Epidemiology*, 64(4):383–394. https://www.sciencedirect.com/science/article/pii/S0895435610003306.

Hanna, J., Joseph, E., Brochhausen, M., and Hogan, W. R. (2013). Building a drug ontology based on RxNorm and other sources. *Journal of Biomedical Semantics*, 4(1):44.

Harnoune, A., Rhanoui, M., Mikram, M., Yousfi, S., Elkaimbillah, Z., and El Asri, B. (2021). BERT based clinical knowledge extraction for biomedical knowledge graph

construction and analysis. *Computer Methods and Programs in Biomedicine Update*, 1:100042.

Hartig, O. (2014). Reconciliation of rdf* and property graphs. https://arxiv.org/abs/1409.3288.

Henry, S., Buchan, K., Filannino, M., Stubbs, A., and Uzuner, O. (2020). 2018 n2c2 shared task on adverse drug events and medication extraction in electronic health records. *Journal of the American Medical Informatics Association*, 27(1):3–12.

Hoelzel, W., Weykamp, C., Jeppsson, J.-O., Miedema, K., Barr, J. R., Goodall, I., Hoshino, T., John, W. G., Kobold, U., Little, R., Mosca, A., Mauri, P., Paroni, R., Susanto, F., Takei, I., Thienpont, L., Umemoto, M., Wiedmeyer, H.-M., and the IFCC Working Group on HbA1c Standardization (2004). IFCC Reference System for Measurement of Hemoglobin A1c in Human Blood and the National Standardization Schemes in the United States, Japan, and Sweden: A Method-Comparison Study. *Clinical Chemistry*, 50(1):166–174.

Hogan, A., Blomqvist, E., Cochez, M., D'amato, C., Melo, G. D., Gutierrez, C., Kirrane, S., Gayo, J. E. L., Navigli, R., Neumaier, S., Ngomo, A.-C. N., Polleres, A., Rashid, S. M., Rula, A., Schmelzeisen, L., Sequeda, J., Staab, S., and Zimmermann, A. (2021). Knowledge graphs. *ACM Computing Surveys*, 54(4):1–37. http://dx.doi.org/10.1145/3447772.

Hogan, A., Dong, X. L., Vrandečić, D., and Weikum, G. (2025). Large language models, knowledge graphs and search engines: A crossroads for answering users' questions. https://arxiv.org/abs/2501.06699.

Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., and Liu, T. (2025). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Trans. Inf. Syst.*, 43(2). https://doi.org/10.1145/3703155.

Imran, S. A., Agarwal, G., Bajaj, H. S., and Ross, S. (2018). Targets for glycemic control. *Canadian Journal of Diabetes*, 42(Supplement 1):S42–S46. https://www.diabetes.ca/for-professionals/full-guidelines/chapter-8.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., tau Yih, W., Rocktäschel, T., Riedel, S., and Kiela, D. (2021). Retrieval-augmented generation for knowledge-intensive nlp tasks. https://arxiv.org/abs/2005.11401.

Li, L., Wang, P., Yan, J., Wang, Y., Li, S., Jiang, J., Sun, Z., Tang, B., Chang, T.-H., Wang, S., and Liu, Y. (2020). Real-world data medical knowledge graph: Construction and applications. *Artificial Intelligence in Medicine*, 103:101817.

Liang, Y., Wang, J., Zhu, H., Wang, L., Qian, W., and Lan, Y. (2023). Prompting Large Language Models with Chain-of-Thought for Few-Shot Knowledge Base Question Generation. In Bouamor, H., Pino, J., and Bali, K., editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 4329–4343, Singapore. Association for Computational Linguistics.

Lieder, H. (2026). Pymupdf/PyMuPDF. PyMuPDF. https://pymupdf.readthedocs.io/en/latest/.

Nasar, Z., Jaffry, S. W., and Malik, M. (2021). Named Entity Recognition and Relation Extraction: State of the Art. *ACM Computing Surveys*, 54.

National Institute for Health and Care Excellence (2015a). Type 1 diabetes in adults: Diagnosis and management. NICE guideline NG17. Last updated 17 August 2022. https://www.nice.org.uk/guidance/ng17.

National Institute for Health and Care Excellence (2015b). Type 2 diabetes in adults: Management. NICE guideline NG28. Last updated 18 February 2026. https://www.nice.org.uk/guidance/ng28.

Nyameino, J. N., Ebbesvik, B.-R., Rabbi, F., Were, M. C., and Lamo, Y. (2020). Model-Driven Automatic Question Generation for a Gamified Clinical Guideline Training System. In Damiani, E., Spanoudakis, G., and Maciaszek, L. A., editors, *Evaluation of Novel Approaches to Software Engineering*, pages 227–245, Cham. Springer International Publishing.

OpenAI, R. (2023). Gpt-4 technical report. arxiv 2303.08774. *View in Article*, 2(5):1.

Peleg, M. (2013). Computer-interpretable clinical guidelines: A methodological review. *Journal of Biomedical Informatics*, 46(4):744–763.

Punthakee, Z., Goldenberg, R., and Katz, P. (2018). Definition, classification and diagnosis of diabetes, prediabetes and metabolic syndrome. *Canadian Journal of Diabetes*, 42(Supplement 1):S10–S15. https://www.diabetes.ca/for-professionals/full-guidelines/chapter-3.

Ramakrishnan, C., Patnia, A., Hovy, E., and Burns, G. (2012). Layout-aware text extraction from full-text PDF of scientific articles. *Source code for biology and medicine*, 7:7.

Savova, G. K., Masanz, J. J., Ogren, P. V., Zheng, J., Sohn, S., Kipper-Schuler, K. C., and Chute, C. G. (2010). Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES): Architecture, component evaluation and applications. *Journal of the American Medical Informatics Association: JAMIA*, 17(5):507–513.

Schreiber, S., Agne, S., Wolf, I., Dengel, A., and Ahmed, S. (2017). Deepdesrt: Deep learning for detection and structure recognition of tables in document images. In *2017 14th IAPR International Conference on Document Analysis and Recognition (ICDAR)*, volume 01, pages 1162–1167.

Singh, B., Afzal, Z., van Mulligen, E. M., and Kors, J. (2013). Using rule-based natural language processing to improve disease normalization in biomedical text. *Journal of the American Medical Informatics Association*, 20:876–881.

Singhal, K., Tu, T., Gottweis, J., Sayres, R., Wulczyn, E., Amin, M., Hou, L., Clark, K., Pfohl, S. R., Cole-Lewis, H., Neal, D., Rashid, Q. M., Schaekermann, M., Wang, A., Dash, D., Chen, J. H., Shah, N. H., Lachgar, S., Mansfield, P. A., Prakash, S., Green, B., Dominowska, E., Agüera y Arcas, B., Tomašev, N., Liu, Y., Wong, R., Semturs, C., Mahdavi, S. S., Barral, J. K., Webster, D. R., Corrado, G. S., Matias, Y., Azizi, S., Karthikesalingam, A., and Natarajan, V. (2025). Toward expert-level medical question answering with large language models. *Nature Medicine*, 31(3):943–950.

Soldaini, L. (2016). Quickumls: a fast, unsupervised approach for medical concept extraction. In *AMIA Annual Symposium Proceedings*. https://api.semanticscholar.org/CorpusID:2990304.

Sousa-Pinto, B., Vieira, R. J., Marques-Cruz, M., Bognanni, A., Gil-Mata, S., Jankin, S., Amaro, J., Pinheiro, L., Mota, M., Giovannini, M., de las Vecillas, L., Pereira, A. M., Lityńska, J., Samolinski, B., Bernstein, J., Dykewicz, M., Hofmann-Apitius, M., Jacobs, M., Papadopoulos, N., Williams, S., Zuberbier, T., Fonseca, J. A., Cruz-Correia, R., Bousquet, J., and Schünemann, H. J. (2024). Artificial Intelligence–Supported Development of Health Guideline Questions. *Annals of Internal Medicine*, 177(11):1518–1529.

Uzuner, Ö., South, B. R., Shen, S., and DuVall, S. L. (2011). 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. *Journal of the American Medical Informatics Association: JAMIA*, 18(5):552–556.

Villena, J., Collada-Pérez, S., Serrano, S., and Gonzalez-Cristobal, J. (2011). Hybrid approach combining machine learning and a rule-based expert system for text categorization. In *Proceedings of the 24th International Florida Artificial Intelligence Research Society, FLAIRS - 24*.

Wei, Q., Ji, Z., Si, Y., Du, J., Wang, J., Tiryaki, F., Wu, S., Tao, C., Roberts, K., and Xu, H. (2020). Relation Extraction from Clinical Narratives Using Pre-trained Language Models. *AMIA Annual Symposium Proceedings*, 2019:1236–1245.

Xu, D., Chen, W., Peng, W., Zhang, C., Xu, T., Zhao, X., Wu, X., Zheng, Y., Wang, Y., and Chen, E. (2024). Large Language Models for Generative Information Extraction: A Survey. arXiv:2312.17617.

Yang, W., Cao, F., and Zhao, X. (2025). Extraction of PDF Table Data Based on the Pdfplumber Method. In *Proceedings of the 2024 4th International Joint Conference on Robotics and Artificial Intelligence*, JCRAI '24, pages 103–106, New York, NY, USA. Association for Computing Machinery.

Yang, Y., Chen, S., Zhu, Y., Liu, X., Ma, W., and Feng, L. (2024). Intelligent extraction of reservoir dispatching information integrating large language model and structured prompts. *Scientific Reports*, 14(1):14140.

Zhao, X., Deng, Y., Yang, M., Wang, L., Zhang, R., Cheng, H., Lam, W., Shen, Y., and Xu, R. (2024). A Comprehensive Survey on Relation Extraction: Recent Advances and New Frontiers. *ACM Computing Surveys*, 56(11):293:1–293:39.

Zheng, X., Burdick, D., Popa, L., Zhong, X., and Wang, N. X. R. (2021). Global Table Extractor (GTE): A Framework for Joint Table Identification and Cell Structure Recognition Using Visual Context. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 697–706.

Zhong, X., ShafieiBavani, E., and Yepes, A. J. (2020). Image-based table recognition: Data, model, and evaluation. arXiv:1911.10683.

A. Prompts and Human Review Material

A.1. LLM Extraction Prompts

The following prompts were used in `llm_extract_relations.py` for schema-constrained entity and relation extraction. The system prompt defines the output schema and extraction rules. The user prompt supplies the guideline identifier, page number, known entities on the page, and the raw text chunk to process.

System Prompt

Listing A.1: LLM extraction system prompt.

```
You are a clinical knowledge graph extractor specialising in
   diabetes guidelines.

Extract ONLY the following entity types from the TEXT:
{entity_type_descriptions}

Extract ONLY the following relation types:
{relation_type_descriptions}

RULES:
- Return STRICT JSON with keys: "entities" (list) and "relations"
    (list).
- Each entity needs: "id" (short unique string within this
   response), "type" (from the list above), "name" (canonical name
   ), "organization" (which guideline this came from).
- Each relation needs: "type" (from list above), "from" (entity
   id), "to" (entity id).
- Drug names: use the most specific name mentioned. If a class is
    mentioned (e.g. SGLT2 inhibitor), create a class entity. If a
   specific drug is mentioned (e.g. empagliflozin), create both.
- Do NOT extract GlycaemicTarget, Recommendation, or numeric
   thresholds -- those are handled separately.
- Do NOT invent relations not explicitly stated in the text.
- If nothing relevant is found, return: {{"entities": [], "
   relations": []}}
- Return ONLY JSON. No markdown fences. No commentary.
```

User Prompt Template

Listing A.2: LLM extraction user prompt template. Placeholders in braces are substituted at runtime.

```
GUIDELINE: {org} ({year})
PAGE: {page}

KNOWN FACTS ON THIS PAGE (already extracted -- do not re-extract
   these):
{known_facts}

TEXT:
{chunk_text}

Return ONLY JSON.
```

A.2. Question Generation Prompts

The following system and user prompts were used in `generate_questions_hybrid.py`. Each generation method has a dedicated builder function that constructs both prompts from the graph anchor at runtime. Placeholders in braces are substituted from the anchor fields.

Graph-Grounded Prompt

Listing A.3: System prompt: Graph-Grounded Prompt.

```
You are a clinical medical education assistant. You write clear,
    accurate questions for healthcare students studying diabetes
   management. Use only the clinical information in the FACT
   block; do not add drug names, thresholds, or guidelines that
   are not stated there. Write the question as a standalone exam
   question. Never refer to 'the fact', 'the provided fact', 'the
    context', 'the passage', or the guideline source inside the
   question. A student reading the question must not be able to
   tell it was generated from a supplied fact.
```

Listing A.4: User prompt template: Graph-Grounded Prompt.

```
{context}

Task: Write ONE clinical {qtype} question that tests knowledge of
    this fact.
{focus_line}- The question must be answerable from the FACT above
- Do not mention page numbers, guideline names, or 'the provided
   fact'
- Write in clear, professional clinical language
- Output ONLY the question text, nothing else
```

Clinical Vignette

Listing A.5: System prompt: Clinical Vignette.

```
You are a clinical medical education assistant writing case-based
    questions for diabetes management training. IMPORTANT: Base
   the clinical scenario ONLY on the facts provided. You may
   describe the patient's condition and symptoms in words, but
   must NOT include any numerical values (no ages, no HbA1c values
   , no glucose readings, no doses) and must NOT introduce drug
   names, targets, or guidelines beyond what is stated in the
   CLINICAL FACT block.
```

Listing A.6: User prompt template: Clinical Vignette.

```
{context}

Task: Write a short clinical vignette question (2-3 sentences).
- Describe a realistic patient with {dst_name} using only words (
   no numbers)
- End with a question about pharmacological management
- The correct answer must be {src_name} (but do not give it away
   in the question)
- Do NOT include any numerical values in the question
- Output ONLY the question text, nothing else
```

Question Reformulation

Listing A.7: System prompt: Question Reformulation.

```
You are a clinical medical education assistant. You rephrase
   questions for different difficulty levels without changing the
   factual content.
```

Listing A.8: User prompt template: Question Reformulation.

```
Original question: "{canonical}"

Task: Rewrite this question at three difficulty levels.
Return ONLY a JSON object with this exact structure:
{{
  "recall":      "direct factual question (easiest)",
  "application": "clinical scenario requiring application of the
     fact",
  "analysis":    "comparative or reasoning question (hardest)"
}}
Do not change any drug names, numerical values, guideline names,
   or clinical facts.
```

MCQ Distractor Generation

Listing A.9: System prompt: MCQ Distractor Generation.

```
You are a clinical medical education expert creating multiple-
   choice questions  for diabetes pharmacology assessment.
```

Listing A.10: User prompt template: MCQ Distractor Generation.

```
Question: {question}
Correct answer: {correct_answer}

Task: Generate exactly 3 wrong but plausible distractor options
   for this multiple-choice question.
Rules:
- Distractors must be real diabetes medications or clinical
   values (choose from: {pool_str})
- Do NOT include "{correct_answer}" as a distractor
- Distractors should be clinically plausible (a student could
   reasonably consider them)
- Return ONLY a JSON object with this exact structure: {{"
   distractors": ["option1", "option2", "option3"]}}
```

A.3. Human Review Question Set 1 (Reviewer: Dr. Aakanksha)

Reviewers were shown the question text only, without the graph anchor, answer key, or source guideline passage.

1. A middle-aged woman with a longstanding history of type 2 diabetes presents to her nephrologist with progressive decline in kidney function, persistent proteinuria, and signs of fluid retention consistent with chronic kidney disease. She has been struggling to achieve adequate glycemic control despite lifestyle modifications, and her physician is considering adding a pharmacological agent specifically indicated for managing chronic kidney disease in patients with her underlying condition. Which patient population is the primary target for this kidney-protective pharmacological management strategy?
2. In a patient undergoing an oral glucose tolerance test, what 2-hour plasma glucose value meets the diagnostic threshold for type 2 diabetes?
3. In the management of type 2 diabetes mellitus, which pharmacological approach involving the combination of agents within a single preparation is recommended as a treatment strategy?
4. In a patient with type 2 diabetes, at what HbA1c threshold would a value be considered above the acceptable glycemic target, indicating suboptimal long-term blood glucose control?
5. How does the 2-hour plasma glucose threshold used to diagnose diabetes on an oral glucose tolerance test according to the American Diabetes Association compare with the threshold recommended by the International Diabetes Federation?
6. In a patient with type 2 diabetes, what is the target fasting plasma glucose level that indicates optimal glycemic control?
7. Which non-pharmacological intervention is recommended as part of the management plan for a patient diagnosed with diabetes?
8. Which medication, a glucagon-like peptide-1 receptor agonist, is recommended for the management of type 2 diabetes mellitus?
9. A middle-aged woman with a longstanding history of type 2 diabetes presents to her general practitioner reporting recurrent episodes of shakiness, sweating, and lightheadedness occurring throughout the week, consistent with hypoglycaemic events. She has no known drug allergies and is currently not on any glucose-lowering therapy. Which oral pharmacological agent is indicated to manage her hypoglycaemic events?
10. In the management of diabetes, which medication is recommended for individuals who are unable to tolerate other pharmacological treatments?
11. A middle-aged man with longstanding type 2 diabetes attends a review clinic at a busy urban practice serving a socioeconomically disadvantaged community. He reports difficulty affording medications, has limited access to healthy food, and faces significant barriers to engaging with lifestyle programmes. Which patient popu-

lation characteristic is particularly relevant when considering the pharmacological management approach for his type 2 diabetes?

12. Which class of medication is recommended for use in adults with type 2 diabetes: SGLT2 inhibitors?
13. Which pharmacological agent is recommended for the management of diabetes in elderly male patients?
14. A middle-aged woman with type 2 diabetes presents to her endocrinologist for a routine follow-up. She has a prior history of ischemic stroke and her physician is considering an oral antidiabetic agent that may offer additional benefit given her cerebrovascular history. Which pharmacological agent would be most appropriate to initiate in this patient to address both her glycemic management and her history of stroke?
15. Which structured programme is recommended for individuals with type 2 diabetes who also have Heart Failure, with the aim of achieving disease remission?

A.4. Human Review Question Set 2 (Reviewer: Dr. Mounika)

Reviewers were shown the question text only, without the graph anchor, answer key, or source guideline passage.

1. A patient with type 1 diabetes mellitus is found to have impaired hypoglycaemic awareness. Based on established clinical recommendations, which intervention is advised for managing this condition?
2. A middle-aged man with type 1 diabetes attends a follow-up appointment reporting persistent postprandial hyperglycemia, particularly after his evening meals, which typically consist of large portions of grilled meat, cheese, and nuts. He describes these meals as very filling and rich, with minimal carbohydrate content but substantial amounts of fat and protein. What pharmacological agent is most appropriate to manage his glycemic response to these high-fat, high-protein mixed meals?
3. A patient with diabetes who sustains a fracture requires adjustment to their antidiabetic regimen. Which antidiabetic agent is recommended for use in the setting of a fracture?
4. Which oral antidiabetic agent is recommended as a pharmacological treatment for children and adolescents diagnosed with type 2 diabetes?
5. Which pharmacological agent is recommended for the management of type 2 diabetes mellitus?
6. Which surgical intervention is recommended as a treatment option for patients with type 2 diabetes?

7. A patient with type 2 diabetes is enrolled in the NHS Path to Remission programme. According to recommendations for this programme, which class of medication is indicated for this patient?
8. A middle-aged woman with a longstanding history of type 2 diabetes presents to her GP reporting recurrent episodes of sweating, trembling, and confusion occurring several times a week, consistent with hypoglycaemic events. She is struggling to manage her condition and her diabetes control has been deteriorating despite ongoing treatment. Which structured programme should be considered to address her hypoglycaemic episodes as part of her overall diabetes management?
9. In the management of hypoglycaemia, which monitoring method is recommended to assess capillary blood glucose levels?
10. In a patient with type 2 diabetes undergoing a 2-hour oral glucose tolerance test, what is the target blood glucose level that should be achieved at the 2-hour mark?
11. A middle-aged patient presents to the emergency department with several days of polyuria, polydipsia, and progressive nausea and vomiting, and is found to have deep, labored breathing along with an altered level of consciousness consistent with diabetic ketoacidosis. Laboratory findings confirm severe metabolic acidosis with elevated ketone bodies in the blood and urine. Which condition-based pharmacological management approach is indicated for this patient's acute presentation?
12. Which class of medication is recommended for use within the NHS Path to Remission programme for the management of type 2 diabetes?
13. Which programme is recommended for individuals who have experienced falls?
14. A middle-aged woman with a history of frequent falls presents to her GP following her third fall in as many months, reporting dizziness and unsteadiness when standing. She has no other significant medical history and her GP is keen to explore structured management options to address her condition and reduce her risk of further falls. Which pharmacological management programme would be most appropriate to initiate for this patient?
15. A patient with type 2 diabetes is found to have an increased cardiovascular risk during routine assessment. Which medication is recommended to help reduce this risk?